Chapter 8

Market Risk Management

8.1 Introduction

Every model in this book so far has focused on estimating a quantity of interest: a price, a return, a probability of default, a forecast. Risk management asks a complementary question: not "what do we expect to happen," but "how bad could it get, and how often." A portfolio manager who forecasts only expected returns is flying blind, since the same expected return can come from a smooth, low-volatility strategy or from one that is usually flat but occasionally loses everything. This chapter introduces the standard toolkit for quantifying and managing that downside for market risk: Value at Risk and Expected Shortfall, maximum drawdown, volatility forecasting, backtesting, liquidity risk, and the model governance and capital adequacy concepts that let a firm and its regulators judge whether it can survive its own worst days. Chapter 9 develops the parallel toolkit for credit risk (probability of default, loss given default, credit scoring), since the two risk types call for different tools.

One worked example runs through most of the chapter: a twenty-day return history for a $10 million equity portfolio, used to compute and compare parametric, historical, Monte Carlo, and GARCH-based risk measures, backtest them, and stress-test the resulting portfolio.

The chapter builds its toolkit in the order a risk desk deploys it: a taxonomy of risk types (Section 8.2.1) to set the scope; Value at Risk (Section 8.2.2), computed three ways and extended to a full portfolio and to a volatility-adjusted version using Chapter 7's GARCH forecasts; and backtesting (Section 8.3.1) to check whether any of it worked out of sample. Later sections extend the VaR machinery in three directions: to options positions via the Greeks (the delta-normal approximation, then the sensitivity-based limits of Section 8.4), to hypothetical crises the historical sample never contained (Section 8.5.1), and, in Section 8.5.3, to a direct test of whether a more flexible machine learning model can out-forecast GARCH.

8.2 Measuring Market Risk

This part builds the standard measures in order of how much they say about the tail, each worked on the same portfolio.

8.2.1 Types of Financial Risk

Suppose a risk manager reports that the fund "lost money because of risk." The statement is true but useless: it does not say whether prices moved against a position, a counterparty failed to pay, or a position could not be sold without moving its price, three different problems calling for three different responses. Financial risk is a family of related concerns, and confusing one type for another is a common and costly mistake. Market risk is the risk of loss from movements in prices, rates, or volatilities that a firm is exposed to through the positions it holds; it is the subject of this chapter. Credit risk is the risk that a counterparty, whether a bond issuer, a loan borrower, or a derivatives counterparty, fails to meet a contractual obligation; Chapter 5 already introduced the pricing consequence of credit risk, and Chapter 9 develops the risk-measurement side of the same problem. Liquidity risk comes in two forms: funding liquidity risk, that a firm cannot meet its cash

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obligations as they come due, and market liquidity risk, that a position cannot be sold quickly without moving its price against the seller, connected to the bid-ask spread and market depth of Chapter 3. Operational risk covers losses from failed internal processes, systems, human error, or external events, including the model risk that arises when a changing environment quietly violates a model's assumptions; Chapter 9 covers a standard capital treatment for it, and Chapter 10 develops the AI methods for detecting and managing it. Underwriting risk, that claims on policies a firm has written exceed the premiums collected, is the insurer's counterpart, and Chapter 11 develops it alongside the reserving and ratemaking machinery it requires. A single adverse event often touches several categories: a large, sudden market move (market risk) can trigger defaults among leveraged counterparties (credit risk) just when many firms are forced to sell into an illiquid market (liquidity risk), the contagion dynamic behind the crises of Chapter 3.

Table 8.1 lists each risk type with its standard measurement tool and where in this book it is developed.

Risk type	Standard measurement tool	Where developed
Market risk	Value at Risk, Expected Shortfall, maximum drawdown	Sections 8.2.2–8.3.1 of this chapter
Credit risk	Probability of default, loss given default, expected loss	Chapter 9
Liquidity risk	Bid-ask spread, liquidity-adjusted VaR	Chapter 3 and later in this chapter
Operational risk	Loss distribution approach, scenario analysis, incident-text and anomaly models	Chapter 10, with the regulatory capital charge in Chapter 9
Underwriting risk	Expected loss ratio, combined ratio, chain-ladder reserves	Chapter 11
Systemic risk	Stress testing, contagion modeling	Chapter 3's financial crises discussion, this chapter's stress-testing section, and Chapter 20's network contagion models

Table 8.1: A taxonomy of financial risk types and their standard measurement tools

8.2.2 Value at Risk: Parametric and Historical Methods

Value at Risk (VaR) answers a deliberately narrow question: over a given holding period and at a given confidence level, what is the loss that will not be exceeded except with some small, specified probability? Formally, the α-level VaR of a portfolio is the smallest loss L such that

$$\Pr(\text{Loss} > L) \leq 1 - \alpha.$$

A 1-day 95% VaR of \$1 million means that, under the model's assumptions, the portfolio is expected to lose more than \$1 million on only about 5% of trading days, or roughly one day in twenty.

Table 8.2 shows twenty days of returns for a hypothetical \$10 million equity portfolio managed by a firm we will call Meridian Capital Partners.

Day	1	2	3	4	5	6	7	8	9	10
Return	0.06%	0.42%	-0.27%	-1.01%	-0.49%	-1.13%	0.13%	1.67%	-0.53%	-0.68%
Day	11	12	13	14	15	16	17	18	19	20
Return	0.65%	0.49%	0.19%	-1.06%	0.02%	0.89%	-1.55%	-0.49%	-2.22%	-1.49%

Table 8.2: Twenty days of returns for a \$10 million equity portfolio

The sample mean return is $\bar{r} = -0.32\%$ and the sample standard deviation is $\hat{\sigma} = 0.937\%$ (with $n-1 = 19$ in the denominator, as in every standard-deviation calculation in this book). The parametric (variance-covariance) method assumes returns are normally distributed and computes VaR directly from the mean,

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standard deviation, and the appropriate normal quantile z_α:

$$\text{VaR}_\alpha = -(\bar{r} - z_\alpha \hat{\sigma}) \times V, \tag{8.1}$$

where V is the portfolio's dollar value. Using $z_{0.95} = 1.645$ and $z_{0.99} = 2.326$ and $V = \$10{,}000{,}000$,

$$\text{VaR}_{95\%} = -(-0.0032 - 1.645(0.00937)) \times 10{,}000{,}000 \approx \$186{,}195,$$

$$\text{VaR}_{99\%} = -(-0.0032 - 2.326(0.00937)) \times 10{,}000{,}000 \approx \$250{,}081.$$

Here, as throughout the chapter, the dollar figures are computed at full precision, with $z_{0.95} = 1.6449$, $z_{0.99} = 2.3263$, and $\hat{\sigma} = 0.9374\%$; the rounded inputs displayed reproduce them to within 0.1%. The historical method makes no distributional assumption: it sorts the observed returns and reads the percentile from the data. Sorting the twenty returns in Table 8.2 from worst to best, the second-worst return (-2.22% and -1.55% are the two worst) corresponds to the 5th percentile of a 20-observation sample, giving

$$\text{VaR}_{95\%}^{\text{historical}} = -(-0.0155) \times 10{,}000{,}000 = \$155{,}000.$$

The two estimates disagree by more than \$30,000, illustrating a persistent tension in risk measurement. The parametric method borrows strength from the whole sample and gives a smooth estimate, but it assumes normality, a poor description of returns with heavier tails, the fat-tails caveat of Chapter 7's GARCH models. The historical method assumes nothing, but with twenty observations it depends heavily on a single data point (here the -1.55% return); in practice it uses one to several years of daily data.

8.2.3 Portfolio VaR with the Variance-Covariance Matrix

The single-series parametric VaR above is a special case of a matrix formula that extends to a portfolio of several assets. Recall the two-asset covariance matrix from Chapter 2's worked AssetA/AssetB example,

$$\Sigma = \begin{pmatrix} 0.000414 & -0.000198 \\ -0.000198 & 0.000118 \end{pmatrix},$$

computed there from the two assets' daily returns, and the equal weights $w = (0.5, 0.5)$ used throughout that chapter. The portfolio's return variance is $w^\top \Sigma w$, the formula Chapter 2 verified numerically, and the parametric VaR of a portfolio worth V dollars, assuming a zero mean return (common in practice, since a one-day expected return is negligible relative to one-day volatility), is

$$\text{VaR}_\alpha \approx z_\alpha \sqrt{w^\top \Sigma w}\, V. \tag{8.2}$$

Applying this to a hypothetical \$2 million position split evenly between AssetA and AssetB, $\sqrt{w^\top \Sigma w} \approx 0.583\%$ (matching the portfolio volatility computed in Chapter 2), so

$$\text{VaR}_{95\%} \approx 1.645(0.00583)(2{,}000{,}000) \approx \$19{,}182,$$

$$\text{VaR}_{99\%} \approx 2.326(0.00583)(2{,}000{,}000) \approx \$27{,}130.$$

This matrix formulation is what scales to real trading books with hundreds or thousands of positions: a risk system maintains the covariance matrix Σ across positions and recomputes $w^\top \Sigma w$ whenever positions change, the pattern Chapter 2 introduced with `weights @ cov_matrix.values @ weights`.

Component VaR: Decomposing Portfolio Risk by Position

A portfolio VaR answers "how much risk does the whole book have," but a risk manager allocating limits across desks, or a portfolio manager deciding which position to trim, needs to know how much each position contributes. Component VaR answers this by decomposing total portfolio variance into per-asset contributions with the same covariance matrix,

$$\text{Component Var}_i = w_i (\Sigma w)_i, \qquad \sum_i \text{Component Var}_i = w^\top \Sigma w, \tag{8.3}$$

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so that each asset's fractional share of total portfolio variance, Component $\text{Var}_i/(w^\top \Sigma w)$, scales the total portfolio VaR into a per-asset Component VaR that sums exactly back to the whole. Applying this to Section 8.2.3's two-asset position, $\Sigma w = (0.000108, -0.00004)$, so

$$\begin{aligned}\text{Component Var}_A &= 0.5(0.000108) = 0.000054,\\ \text{Component Var}_B &= 0.5(-0.00004) = -0.00002,\end{aligned}$$

summing to the portfolio variance of 0.000034 exactly as required. Scaling these fractions (1.588 and −0.588) by the total $\text{VaR}_{95\%}$ of \$19,182 gives

$$\text{Component VaR}_A \approx \$30{,}466, \qquad \text{Component VaR}_B \approx -\$11{,}284,$$

which sum back to the \$19,182 total, as the decomposition guarantees. AssetB's component is *negative*, although its standalone volatility is positive. Its strongly negative correlation with AssetA (Section 8.5.1 computes $\rho \approx -0.896$) makes it a diversifier in this portfolio: adding more AssetB, holding AssetA fixed, would *reduce* total risk, which the standalone volatilities could never reveal. That is why trading firms set and monitor risk limits with component or marginal VaR rather than standalone size or volatility: a position's risk contribution depends on how it interacts with the rest of the book.

8.2.4 Expected Shortfall: Going Beyond VaR

VaR has a well-known weakness: it answers "how much can I lose with 95% confidence" but says nothing about how bad the remaining 5% of outcomes are. Two portfolios can have identical VaR while one has a slightly worse tail and the other a catastrophically worse one. Expected Shortfall (ES), also called Conditional VaR, averages the losses in the tail beyond VaR rather than reporting its boundary:

$$\text{ES}_\alpha = \mathbb{E}[\text{Loss} \mid \text{Loss} > \text{VaR}_\alpha]. \tag{8.4}$$

For a continuous loss distribution it makes no difference whether this inequality is strict ($>$) or not ($\geq$). In a finite sample it does, since the VaR is itself one of the observed losses, so a discrete convention is needed: this book counts the VaR observation itself as part of the tail, averaging every loss at or beyond the VaR (other texts average only the $n(1-\alpha)$ losses strictly beyond it, which with twenty observations at 95% would leave just the single worst day). For the historical method applied to Table 8.2, the two worst returns (−2.22% and −1.55%) are the ones at or beyond the 95% VaR threshold, so

$$\text{ES}_{95\%}^{\text{historical}} = -\frac{(-0.0222) + (-0.0155)}{2} \times 10{,}000{,}000 = \$188{,}500.$$

The parametric Expected Shortfall under normality has a closed-form expression involving the standard normal density $\phi(\cdot)$,

$$\text{ES}_\alpha = \left(\hat{\sigma}\frac{\phi(z_\alpha)}{1-\alpha} - \bar{r}\right) \times V, \tag{8.5}$$

which for $\alpha = 95\%$ gives $\text{ES}_{95\%} \approx \$225{,}366$, larger than the parametric $\text{VaR}_{95\%}$ of \$186,195, as it must be, since ES averages over the tail rather than reporting its edge. Because ES is a coherent risk measure in a precise sense that VaR is not (it satisfies subadditivity, so diversification can never increase measured risk, which VaR can violate in pathological cases), regulators have shifted market-risk capital from VaR toward Expected Shortfall, a change taken up in this chapter's discussion of regulatory capital. Figure 8.1 shows why the two measures differ: VaR marks only the boundary of the shaded loss tail, while ES averages over the entire tail beyond it.

8.2.5 Maximum Drawdown: A Path-Dependent Risk Measure

VaR and Expected Shortfall summarize the distribution of one-day losses, but neither says how much value an investor would have seen erode from its peak over an extended period. Maximum drawdown answers

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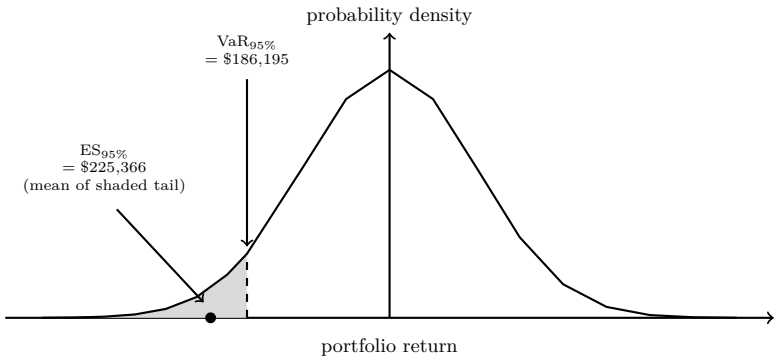


Figure 8.1: The parametric VaR and Expected Shortfall of Table 8.2's portfolio, illustrated against the assumed normal return distribution

this. Given a portfolio value path $V_0, V_1, \ldots, V_T$, the drawdown at time t is the percentage decline from the running peak up to and including that point,

$$\text{Drawdown}_t = \frac{\max_{0\le s\le t} V_s - V_t}{\max_{0\le s\le t} V_s} \ge 0,$$

and the maximum drawdown is the largest drawdown along the path, $\text{MDD} = \max_{0\le t\le T} \text{Drawdown}_t$. Unlike VaR and Expected Shortfall, which depend on the distribution of returns and not their order, maximum drawdown is path-dependent: shuffling the same twenty returns would generally change it while leaving every VaR and ES figure unchanged, because the peak and the trough depend on *when* the losses occur, not just how large they are.

Applying this to the cumulative value path implied by Table 8.2's twenty daily returns, starting from the portfolio's initial \$10,000,000, the running peak of \$10,048,025 is reached after Day 2, following two small positive returns, and is never exceeded again; the lowest point, \$9,371,210, comes at the end, Day 20, after four consecutive negative returns on Days 17 through 20. The maximum drawdown is therefore

$$\text{MDD} = \frac{10{,}048{,}025 - 9{,}371{,}210}{10{,}048{,}025} \approx 6.74\%, \qquad \text{a peak-to-trough decline of \$676,815.}$$

Figure 8.2 plots the path, its running peak, and the peak-to-trough decline. This differs materially from the cumulative twenty-day return of only -6.29%. The extra 0.45 percentage points reflect the portfolio's being briefly *up* \$48,025 before its losses began, so an investor holding from Day 2 onward saw a larger decline than a start-to-end comparison reveals. Maximum drawdown thus captures something daily VaR cannot: a portfolio whose daily losses all clear a well-calibrated 95% VaR threshold can still suffer a large, sustained drawdown if enough of them come consecutively, as in this example's final four days.

Maximum drawdown is used wherever a full return path is the object of interest; Chapter 13 uses it, alongside the Sharpe ratio and hit rate, to evaluate backtested strategies. Asset managers and hedge funds report it routinely because, unlike VaR, it needs no distributional assumption, only a value path, and it answers an investor's plain question: "how much could I have lost had I entered at the worst possible time?" Its main limitation mirrors that simplicity: a realized maximum drawdown is one observation of a path-dependent statistic, not an estimate of a distribution, so comparing two strategies' historical drawdowns says less about future worst-case risk than comparing their VaR or Expected Shortfall.

8.2.6 Monte Carlo Simulation for Risk Measurement

Monte Carlo simulation sits between the two: it assumes a distributional model, as the parametric method does, but instead of a closed-form formula it simulates many return scenarios from the model and reads VaR and ES off the simulated distribution, as the historical method reads them off the observed one. Simulating 100,000 draws from a normal distribution with the same mean and standard deviation as Table 8.2 ($\bar{r} = -0.32\%$, $\hat{\sigma} = 0.937\%$) gives $\text{VaR}_{95\%} \approx \$187{,}460$, $\text{VaR}_{99\%} \approx \$251{,}690$, and $\text{ES}_{95\%} \approx \$226{,}378$. All are

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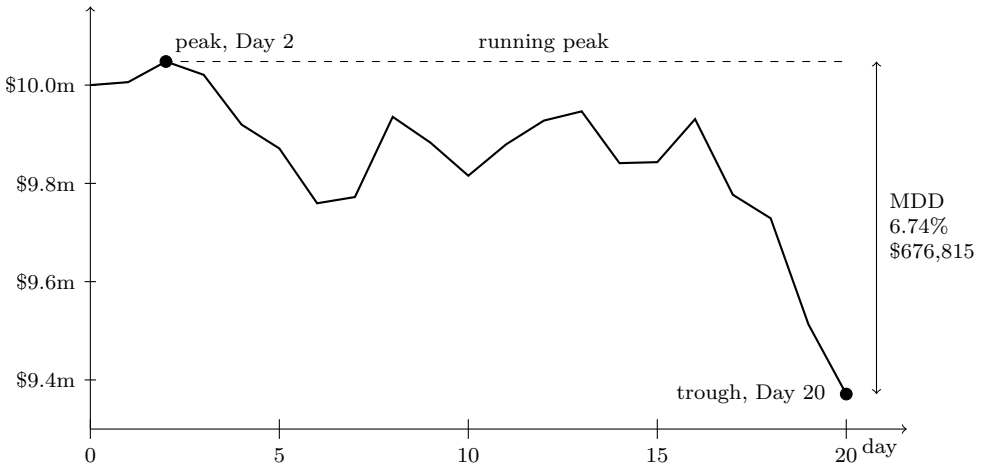


Figure 8.2: The value path of Table 8.2's portfolio, its running peak, and its maximum drawdown

close to the parametric formulas, as they should be: with enough draws, Monte Carlo VaR converges to the parametric answer whenever a closed form exists.

The method's value appears when no closed form is available, for example for a portfolio containing options (nonlinear in the underlying, a case taken up in the delta-normal discussion below) or a model with fat tails, jumps, or the time-varying volatility of Chapter 7's GARCH models. Monte Carlo remains simple there, simulating many paths under the more realistic model and reading off the empirical percentile, while a closed form either does not exist or is much harder to derive.

8.2.7 Volatility-Adjusted VaR: Connecting GARCH Forecasts to Risk Measurement

Every VaR method above uses a single volatility estimate, computed from a fixed historical window and held fixed. Because volatility clusters and evolves, Chapter 7's GARCH(1,1) model suggests an alternative: update VaR every day with the model's one-step-ahead variance forecast, an approach widely used in practice and sometimes called volatility-adjusted or "GARCH VaR."

Recall Chapter 7's GARCH(1,1) recursion applied to Chapter 2's four AssetA returns ($\omega = 0.00002$, $\alpha = 0.10$, $\beta = 0.85$): the one-step-ahead forecast volatility was 1.95% immediately before the large +2.97% return at $t = 3$, and rose to 2.08% immediately after it. Applying the same parametric VaR formula as Section 8.2.2, but with each of these two GARCH-forecast volatilities in place of a static historical standard deviation, to a hypothetical $10 million position gives

$$\text{VaR}_{95\%}^{\text{pre-shock}} = 1.645(0.0195)(10{,}000{,}000) \approx \$320{,}775,$$
$$\text{VaR}_{95\%}^{\text{post-shock}} = 1.645(0.0208)(10{,}000{,}000) \approx \$342{,}160,$$

Full precision ($z_{0.95} = 1.64485$) gives $320,746 and $342,130, the values the notebook prints; the rounded z is shown for readability. an increase of about 6.7% in a single day from the updated variance forecast alone, with no change to the portfolio. A VaR based on a rolling historical standard deviation would still be averaging in the calmer pre-shock returns and would understate the risk while they remained in the window, the lag GARCH's recursive weighting avoids.

Filtered Historical Simulation: Combining the Two Methods

The historical method and the GARCH-based method solve different problems well: the historical method captures the shape of the return distribution, its skew and fat tails, without assuming normality, while GARCH captures how far its width currently sits above or below its long-run level. Filtered historical simulation combines the two. Each historical return r_t is standardized by the GARCH volatility prevailing

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that day, $z_t = r_t/\sigma_t$, giving residuals roughly free of volatility clustering if the GARCH model is well specified; VaR is read off a percentile of these residuals and rescaled by *today's* GARCH forecast rather than the volatility prevailing when each return was observed.

With the two forecasts already computed, $\sigma_3 = 1.95\%$ (pre-shock) and $\sigma_4 = 2.08\%$ (post-shock), a large return observed in the calmer $t = 3$ regime, divided by that day's small σ_t, gives a large standardized residual, as it should, since the return was large *relative to its own regime*.

Chapter 2's AssetA series supplies such an observation. The +2.97% return arrived while $\sigma_3 = 1.95\%$ prevailed, the calmest forecast in the whole recursion, so its standardized residual is

$$z_3 = \frac{0.0297}{0.0195} = 1.523,$$

and rescaling that residual by the post-shock forecast $\sigma_4 = 2.08\%$ gives a hypothetical return for today's conditions of $1.523 \times 0.0208 = 3.17\%$. Plain historical simulation would have reused the 2.97% unchanged, understating the move by about twenty basis points only because the return was observed in a calmer regime than today's.

Rescaling standardized residuals by today's forecast in this way avoids both the plain historical method's problem (treating every past return as if it occurred under today's volatility) and the parametric method's (imposing normality on a shape the data may contradict). Filtered historical simulation is a practical middle ground used in production VaR systems at large trading desks, applying the same GARCH machinery across a whole return history.

8.2.8 Fat Tails: Student-t VaR and Extreme Value Theory

Filtered historical simulation lets the data's shape into the calculation; the more common fix replaces the normal distribution in the parametric formula with a heavier-tailed one. The standard choice is Student's t distribution with ν degrees of freedom, whose density decays like a power $|x|^{-(\nu+1)}$ rather than like the normal's $e^{-x^2/2}$, so that a five- or ten-standard-deviation day is rare but not, as under normality, effectively impossible. The smaller ν, the heavier the tails; as $\nu \to \infty$ the t becomes the normal.

The first question is what ν to use, and Table 8.2 cannot answer it. Maximizing the t likelihood over ν on the twenty returns sends the estimate off toward infinity, back to the normal, and the sample's excess kurtosis is -0.007, no fatter than a normal sample. The likelihood is also nearly flat: at $\nu = 4$ it is only 0.68 below its maximum, so the data are equally consistent with thin and with quite heavy tails. Twenty days contain almost no information about a tail that shows itself a few times a year. The tail parameter has to come from a long history or from what is known about the asset class, and estimates on long histories of daily equity returns typically fall between about 3 and 6 (McNeil et al. 2015). Take $\nu = 4$.

The scaling is where errors creep in. A standard t_ν variable has variance $\nu/(\nu-2)$, which is 2 at $\nu = 4$, so plugging its quantile into Equation (8.1) next to $\hat{\sigma}$ would double-count the spread. The t must first be rescaled to have the sample's standard deviation, with scale $s = \hat{\sigma}\sqrt{(\nu-2)/\nu} = 0.937\% \times \sqrt{1/2} = 0.663\%$. Then, with $t_{\nu,\alpha}$ the t quantile and f_ν its density,

$$\mathrm{VaR}_\alpha = -(\bar{r} - t_{\nu,\alpha}\, s) \times V, \qquad \mathrm{ES}_\alpha = \left(s\, \frac{f_\nu(t_{\nu,\alpha})}{1-\alpha}\, \frac{\nu + t_{\nu,\alpha}^2}{\nu - 1} - \bar{r} \right) \times V, \tag{8.6}$$

the t counterparts of Equations (8.1) and (8.5). Table 8.3 compares the two models at three confidence levels.

Confidence	Normal VaR	Student-t VaR	Normal ES	Student-t ES
95%	\$186,195	\$173,313	\$225,366	\$244,308
99%	\$250,081	\$280,373	\$281,847	\$378,056
99.9%	\$321,690	\$507,488	\$347,644	\$674,069

Table 8.3: Parametric VaR and Expected Shortfall of the \$10 million portfolio under the normal and under a Student-t with $\nu = 4$, both with the sample mean and standard deviation

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Two features matter more than the individual numbers. First, fat tails do not raise every risk figure. Holding the standard deviation fixed, the t puts more probability both very near the mean and very far from it, and less in the shoulders, so its 95% VaR is *lower* than the normal's; the two VaR curves cross at a confidence level of 97.47%, and only beyond it does the t report more risk. A desk that switches to a fat-tailed model and checks only its 95% VaR can wrongly conclude that the change made it safer. Second, the gap widens rapidly in the tail: at 99.9% the t VaR is 58% above the normal's, and the t Expected Shortfall nearly double. Expected Shortfall is more sensitive to the tail model than VaR, the price of being the measure that looks into the tail.

Extreme value theory. The Student-t still imposes one shape on the whole distribution, so its tail is calibrated mostly by the ordinary days in the middle. Extreme value theory models the tail on its own. Its central result, the Pickands-Balkema-de Haan theorem, says that for a wide class of distributions the excesses of losses over a high threshold u are approximately distributed as a *generalized Pareto distribution* (GPD) with a shape ξ and a scale β, whatever the distribution looks like in the middle. A positive ξ means a power-law tail ($\xi = 1/\nu$ for a Student-t), and $\xi = 0$ an exponential, normal-like one. Fitting the GPD to the N_u losses above u out of n observations gives tail estimates

$$\mathrm{VaR}_q = u + \frac{\beta}{\xi}\left[\left(\frac{n}{N_u}(1-q)\right)^{-\xi} - 1\right], \qquad \mathrm{ES}_q = \frac{\mathrm{VaR}_q + \beta - \xi u}{1-\xi},$$

which extrapolate beyond the largest loss observed, something historical simulation can never do. This *peaks-over-threshold* method needs many exceedances, so it cannot be run on twenty days either. Simulating ten years, 2,500 days, from the $\nu = 4$ model above and setting u at the 95th percentile of losses (1.758%, with 125 exceedances) gives $\hat{\xi} = 0.013$ and $\hat{\beta} = 0.657\%$, and a 99.9% VaR of \$439,419 against a true value of \$507,488 and a normal fitted to the same sample of \$321,071. The method recovers almost two-thirds of the gap the normal leaves, but the shape estimate is far below the true $\xi = 0.25$, with a rough standard error of 0.09: even 125 exceedances pin down a tail index only loosely. That trade-off, a model built for the tail against scarce tail data, is why extreme value theory is used where the quantile is extreme and the history long, for 99.9% economic capital, for the severity distributions of Chapter 10's operational losses, and as a check on the 97.5% Expected Shortfall that now sets market-risk capital.

8.3 Validating and Extending the Measures

A risk number nobody has tested is a decoration; this part checks the measures against realized losses and extends them where the simple versions break.

8.3.1 Backtesting Value at Risk Models

A VaR model is useful only if it is calibrated, and the standard check is backtesting: comparing how often realized losses exceeded the VaR threshold (an "exception") with how often the model predicted they would. A well-calibrated 95% VaR model should produce exceptions on roughly 5% of days, no more and no fewer, since a model that is never exceeded may be too conservative, tying up capital that could be deployed productively.

Applying the parametric 95% VaR threshold from Section 8.2.2 (a daily return of -1.862%, equivalent to the \$186,195 VaR figure) to the twenty returns in Table 8.2, exactly one return, the -2.22% observation, falls below the threshold, for an observed exception rate of $1/20 = 5\%$, matching the model's target almost exactly. Applying the parametric 99% threshold (a return of -2.501%) produces zero exceptions, the natural outcome for an expected count of $0.01 \times 20 = 0.2$.

Kupiec's proportion-of-failures (POF) test formalizes this comparison, testing the null hypothesis that the model's true exception rate equals its target p against a likelihood-ratio statistic,

$$LR_{\mathrm{POF}} = -2\ln\left[\frac{(1-p)^{n-x}p^x}{(1-x/n)^{n-x}(x/n)^x}\right], \tag{8.7}$$

which is compared against a chi-squared distribution with one degree of freedom (critical value 3.841 at the conventional 95% test level) under the null. For the twenty-day sample above, with $p = 0.05$, $n = 20$, and

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$x = 1$ exception, the observed rate $x/n = 1/20 = 0.05$ exactly equals the target p, so the numerator and denominator are identical and $LR_{\text{POF}} = 0$, nowhere close to rejecting the null.

This looks like a perfect endorsement but is really a small-sample caution: one more bad day would move the rate to $2/20 = 10\%$, and even that doubled rate would not, on twenty observations, distinguish a well-calibrated model from a miscalibrated one. Kupiec's test, and the Basel traffic-light framework built on the same logic, need a much longer history, typically the one-year, 250-observation window described below, before they have real power. Like Chapter 7's Dickey-Fuller test, a test can be correctly constructed and still lack the power to detect a problem in a short sample.

Regulators take backtesting failures seriously: under the Basel framework, a bank whose internal VaR model produces too many exceptions in a rolling year can be required to hold more regulatory capital through a multiplier on the reported VaR, giving banks a financial incentive to calibrate their models rather than report an artificially low VaR.

The Basel Traffic-Light Backtesting Framework

Basel formalizes this backtesting logic into a three-zone "traffic light" system, based on the number of VaR exceptions observed over a rolling 250-trading-day window at the 99% confidence level, where $0.01 \times 250 = 2.5$ exceptions are expected on average if the model is correctly calibrated. Zero to four exceptions fall in the green zone, requiring no action beyond continued monitoring and leaving the baseline capital multiplier of 3 unchanged. Five to nine exceptions fall in the yellow zone, triggering closer scrutiny and a multiplier that rises within the zone; ten or more fall in the red zone, where the model is presumed flawed and the multiplier reaches its maximum of 4.

The framework works only over its full 250-day window: Table 8.2's twenty-day, one-exception sample cannot be classified into a zone, since the thresholds are calibrated to 250 observations, not twenty. Naively extrapolating the observed rate ($1/20 = 5\%$) to a year would suggest twelve or thirteen exceptions, well into the red zone, but that repeats the small-sample overreach just cautioned against. The traffic-light system is designed to be read only once a full 250-day window has been observed.

Backtesting Expected Shortfall

The shift from VaR to Expected Shortfall complicates backtesting. VaR backtesting reduces to counting exceptions, a binary event on which Kupiec's test and the traffic-light framework build, but ES is an average of losses *within* the tail, so a single exceedance does not say whether the ES forecast was too high, too low, or about right.

A common approach compares the *realized* loss on each exception day with the model's ES forecast for that day, averaged over the exceptions. Table 8.2's single 95% exception, the -2.22% return, realizes a loss of $0.0222 \times 10{,}000{,}000 = \$222{,}000$ against the parametric $\text{ES}_{95\%}$ forecast of \$225,366 computed earlier, a ratio of about 0.985, suggesting the forecast was, if anything, slightly conservative on this occasion.

One exception is far too little evidence to validate an ES model, but this comparison, accumulated over many exception days, is how a trading desk builds the track record needed to judge whether its ES model is calibrated, not merely whether its VaR threshold is crossed the right number of times.

8.3.2 From Daily VaR to Economic Capital: Time Scaling and Its Limits

Every VaR figure above answers a one-day question, but two related questions follow in practice: how much capital to hold against a longer regulatory horizon, and how much the institution as a whole needs over a year to survive a severe loss with high confidence. Both build on the daily VaR, through a time-scaling shortcut and a much higher confidence level respectively.

Basel's market-risk capital framework historically scaled a bank's daily VaR up to a 10-day regulatory horizon using the square-root-of-time rule,

$$\text{VaR}_T \approx \text{VaR}_1 \sqrt{T}, \tag{8.8}$$

which holds exactly when daily returns are independent and identically distributed, since the variance of a sum of T such returns is T times the one-day variance. Applied to the parametric 99% VaR of \$250,081 from Section 8.2.2, it gives a 10-day regulatory VaR of $250{,}081\sqrt{10} \approx \$790{,}824$ without a separate 10-day model.

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The independence assumption, however, is the one Chapter 7 warned against: daily returns are rarely independent, and even modest positive autocorrelation changes how fast multi-day risk grows. If daily returns followed an AR(1) process with $\phi = 0.10$ (much weaker than the $\phi = 0.71$ Chapter 7 estimated for interest-rate deviations, but realistic for daily equity returns), the variance of a T-day cumulative return is

$$\text{Var}\left(\sum_{t=1}^{T} r_t\right) = T\sigma^2 + 2\sigma^2 \sum_{k=1}^{T-1} (T-k)\phi^k,$$

which for $T = 10$ gives a 10-day volatility about 9.4% higher than the square-root-of-time rule assumes, and a 10-day 99% VaR of about \$865,413 rather than \$790,824: applied to autocorrelated returns, the shortcut understates the capital a bank should hold, in the direction that matters.

Economic capital takes the logic one step further, from a 10-day horizon to a one-year horizon at a much higher confidence level, typically 99.9% or more, consistent with the default probability of the rating (commonly single-A or double-A) a bank wants its solvency to reflect; Chapter 9 develops the analogous calculation for credit portfolios with the Vasicek model. Unlike the regulatory VaR multiplier of Section 8.3.1, a supervisory charge computed by a prescribed method, economic capital is the bank's own estimate of the capital its risk-taking needs to survive a severe, low-probability year. It serves internal capital allocation and performance measurement as much as compliance: a desk earning a given return on a smaller economic-capital charge uses the firm's scarce capital more efficiently.

8.3.3 Liquidity Risk and Liquidity-Adjusted VaR

Every VaR calculation so far has assumed a position can be liquidated at the market price, but, as Chapter 3 showed, selling costs at least half the bid-ask spread relative to the mid price, and a large order can walk through several levels of the order book at progressively worse prices. Liquidity-adjusted VaR (LVaR) adds an estimated liquidation cost to VaR,

$$\text{LVaR}_\alpha = \text{VaR}_\alpha + \frac{1}{2}\,(\text{spread}) \times V,$$

where the spread is a fraction of position value and the factor of one-half is the cost of crossing from the mid price to the relevant side of the quote, as in Chapter 3's round-trip cost. Applying this to the \$2 million two-asset position from Section 8.2.3, assuming an average bid-ask spread of 30 basis points, the liquidity cost adjustment is

$$\frac{1}{2}(0.0030)(2{,}000{,}000) = \$3{,}000, \qquad \text{LVaR}_{95\%} = 19{,}182 + 3{,}000 = \$22{,}182,$$

an increase of roughly 16% over the standard parametric VaR of \$19,182. The adjustment matters most where it is most often ignored. In a market-wide liquidity crunch, spreads widen sharply, since they compensate market makers for inventory risk and adverse selection, both of which rise under stress. The cost of liquidating just when a firm most needs to reduce risk can therefore far exceed an adjustment calibrated on calm-market spreads, an instance of the procyclicality problem discussed later in this chapter.

Market liquidity risk, the focus above, is distinct from funding liquidity risk, the risk that a firm cannot meet its own near-term cash obligations, and Basel III addresses the latter directly with the Liquidity Coverage Ratio (LCR),

$$\text{LCR} = \frac{\text{High-Quality Liquid Assets}}{\text{Total Net Cash Outflows over 30 days}} \geq 100\%,$$

which requires a bank to hold enough liquid assets (broadly, cash and high-grade government bonds) to survive 30 days of a specified stress scenario without raising new funding. A bank with \$50 million in high-quality liquid assets and projected net cash outflows of \$40 million over a 30-day stress scenario has $\text{LCR} = 50/40 = 125\%$, comfortably above the regulatory minimum. The LCR is the funding-side complement to liquidity-adjusted VaR: LVaR asks what a position would cost to sell, the LCR whether the firm has enough liquid resources to avoid being a forced seller at all during stress. Forced selling into a stressed,

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illiquid market is what turned market liquidity risk into a systemic problem in the 2008 crisis discussed later in this chapter.

Basel III pairs the LCR with a second, longer-horizon ratio, the Net Stable Funding Ratio (NSFR),

$$\text{NSFR} = \frac{\text{Available Stable Funding}}{\text{Required Stable Funding}} \geq 100\%,$$

which asks the LCR's question over a one-year horizon rather than 30 days, weighted toward the structural mismatch between how long a bank's assets take to mature (loans, often years) and how quickly its liabilities could be withdrawn (deposits, often on demand). Available stable funding weights a bank's capital and liabilities by how reliably they will still be there in a year (retail deposits are weighted as more stable than short-term wholesale borrowing, for instance), while required stable funding weights its assets by how long they are locked up (a long-dated loan requires more stable funding backing it than an overnight reverse repo). Where the LCR addresses a sudden, short-term liquidity run, the NSFR addresses the slower-moving risk of a bank funding long-term assets with short-term, flighty liabilities, the maturity-transformation mismatch that turned a funding problem into a solvency problem for several institutions during the 2008 crisis once wholesale funding markets froze.

8.3.4 VaR for Nonlinear Portfolios: The Delta-Normal Approximation and Its Limits

Every VaR calculation so far has assumed a linear portfolio, whose value moves in constant proportion to the risk factor. A portfolio containing options violates this, and Chapter 5's Greeks show how.

Consider a market maker short 1,000 of the one-year, \$50-strike calls priced in Chapter 5 ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $C_0 \approx \$6.88$, $\Delta = 0.6115$, $\Gamma = 0.0255$). The delta-normal method approximates the position's one-day loss using only delta, treating the option price as locally linear in the stock price. Converting the annual volatility to a daily figure the standard way, $\sigma_{\text{daily}} = 0.30/\sqrt{252} \approx 1.90\%$, the position's delta-implied one-day dollar standard deviation is

$$\Delta \times S_0 \times \sigma_{\text{daily}} \times 1{,}000 = 0.6115 \times 50 \times 0.0190 \times 1{,}000 \approx \$577.85,$$

giving $\text{VaR}_{95\%}^{\text{delta-normal}} \approx 1.645(577.85) \approx \$950.56$ and $\text{VaR}_{99\%}^{\text{delta-normal}} \approx 2.326(577.85) \approx \$1{,}344.08$.

Full revaluation instead reprices the same option with the same Black-Scholes-Merton formula at a stressed stock price, holding strike, rate, volatility, and time to maturity fixed (a simplification that ignores one day of time decay, isolating the delta and gamma effect this example is meant to illustrate). The worst case for a short call is the stock rising, so at the 95% stressed price $S_0(1 + z_{0.95}\sigma_{\text{daily}}) \approx \$51.55$, the same option reprices to \$7.86, a position loss of \$980.79, about 3.2% *more* than the delta-normal estimate. At the more extreme 99% stressed price of \$52.20, the option reprices to \$8.28, a loss of \$1,403.99, now about 4.5% more than delta-normal predicted.

A delta-gamma approximation closes most of this gap without requiring a full revaluation, by adding a quadratic correction term built directly from the same gamma already computed in Chapter 5,

$$\Delta P \approx \left(\Delta\,\Delta S + \tfrac{1}{2}\Gamma\,\Delta S^2\right) \times n_{\text{contracts}}. \tag{8.9}$$

With $\Delta S_{95} = S_0 z_{0.95}\sigma_{\text{daily}} \approx \$1.554$ and $\Delta S_{99} \approx \$2.198$, this gives delta-gamma losses of

$$1{,}000\left(0.6115(1.554) + \tfrac{1}{2}(0.0255)(1.554)^2\right) \approx \$981.43,$$
$$1{,}000\left(0.6115(2.198) + \tfrac{1}{2}(0.0255)(2.198)^2\right) \approx \$1{,}405.79,$$

within a few cents of the full-revaluation figures of \$980.79 and \$1,403.99, compared to delta-normal's \$950.56 and \$1,344.08. The quadratic term costs one extra number, gamma, and recovers nearly all the accuracy of full repricing, which is why delta-gamma VaR is the standard compromise for portfolios with too many options to reprice at every stressed scenario. The growing gap is gamma's effect: delta is the option's slope at the current price, but a large move travels far enough along the price-value curve that the slope has changed by the time it is complete, and gamma measures that curvature. Delta-normal is a cheap, reasonable approximation for small moves or little gamma exposure, but it understates risk for a portfolio with meaningful gamma, and more so the larger the move, which is why the 99% gap above is proportionally wider than the 95% gap.

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8.3.5 Interest-Rate Risk in the Banking Book

Every position measured so far sits in a *trading book*: held to be sold or hedged within days, marked to market every evening, its risk expressed over a one-day or ten-day horizon. Most of a commercial bank's balance sheet is different. Its loans, its deposits, and the large securities portfolio it buys with the deposits it has not lent out make up the *banking book*, which is held for the long term and accounted for largely at amortized cost. Securities a bank designates as *held to maturity* (HTM) are carried at what the bank paid for them, adjusted for amortization, whatever they would fetch today. Securities *available for sale* (AFS) are carried at fair value, but in the United States a bank outside the largest size categories may elect to keep their unrealized gains and losses out of its regulatory capital. A rise in rates that takes 18% off the market value of a bond portfolio can therefore leave both reported earnings and regulatory capital almost untouched.

The loss is real nonetheless. A bond worth \$82 today is worth \$82 whoever owns it; the accounting label decides only when the loss is recognized. If the bank is never forced to sell, the bond pulls back to par and the bank recovers its book value, but only by earning a below-market coupon for the remaining years: the loss arrives as thin interest margins instead of one write-down. Holding to maturity is a bet that the bank will never need the cash first, a bet about its depositors. Measuring interest-rate risk in the banking book (IRRBB) means measuring that exposure whatever the accounting says, in two complementary ways.

Economic value of equity. The first measure asks what the balance sheet is worth today, the present value of the asset cash flows minus the present value of the liability cash flows, and how much that difference moves when rates move. Applying Chapter 5's duration approximation to each side of the balance sheet gives

$$\Delta \mathrm{EVE} \approx -\Big(\sum_i D_i A_i - D_L L \Big)\, \Delta y, \tag{8.10}$$

where A_i and D_i are the value and modified duration of each class of assets, and L and D_L are the value and effective duration of the liabilities. A bank whose assets have a longer duration than its liabilities, the normal condition of an institution funding long-term loans and bonds with demand deposits, loses economic value when rates rise. The awkward term is D_L: a demand deposit has no contractual maturity, so its duration is a behavioral assumption about how long depositors will leave their money, estimated from the bank's own history.

Net interest income. The second measure asks what happens to earnings over the next year. Assets that reprice within the year, cash and floating-rate loans, start earning more at once; fixed-rate bonds do not. On the other side, banks raise deposit rates only partially and slowly, and the fraction of a market-rate move that is passed through to depositors is the *deposit beta* β. To first order,

$$\Delta \mathrm{NII} \approx \big(A_{\mathrm{reprice}} - \beta\, L_{\mathrm{dep}} \big)\, \Delta y, \tag{8.11}$$

with A_{reprice} the assets that reprice within the horizon and L_{dep} the deposits. Drechsler et al. (2017) show that low deposit betas reflect banks' market power over depositors, which is why banks treat deposits as cheap, stable, long-duration funding that hedges the duration of their assets. The two measures can point in opposite directions: a bank can be *asset sensitive*, with net interest income that rises when rates rise, while its economic value falls, because NII sees only the next twelve months and EVE sees all of them.

The supervisory standard. The Basel Committee's IRRBB standard of April 2016 (BCBS 2016a) keeps the topic in Pillar 2, the supervisory-review pillar, rather than imposing a mechanical capital charge. It prescribes six standardized rate-shock scenarios (parallel up, parallel down, steepener, flattener, short rates up, and short rates down; for the U.S. dollar the 2016 calibration sets the parallel shock at 200 basis points) and a *supervisory outlier test*: a bank whose largest decline in EVE across those scenarios exceeds 15% of its Tier 1 capital is flagged as an outlier, which can lead supervisors to require it to reduce the risk or hold more capital. Because the test runs on EVE, it inherits Equation (8.10)'s dependence on the deposit-duration assumption, as the worked example shows.

Worked Example 8.1: A Deposit-Funded Bank Meets a 300-Basis-Point Rise

Consider a stylized bank with \$100 billion of assets: \$8 billion of cash, \$42 billion of floating-rate loans that reset every quarter (modified duration about 0.25), and a \$50 billion securities portfolio of seven-year bonds with a 1.50% annual coupon, bought at par when seven-year yields were 1.50%. It is funded by \$91 billion of

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deposits and \$9 billion of equity, an ordinary leverage for a bank. Chapter 5's formulas give the bonds a Macaulay duration of 6.697 years and a modified duration of $D_{\text{mod}} = 6.697/1.015 = 6.598$. Repricing them at yields of 2.50% and 0.50% gives \$93.65 and \$106.86 per \$100 of par, so Chapter 5's three-price estimate of convexity is

$$\text{Convexity} \approx \frac{93.65 + 106.86 - 2(100)}{100\,(0.01)^2} \approx 51.27.$$

Now let yields rise 300 basis points, to 4.50%, a move of roughly the size seen in 2022. The duration approximation predicts a price change of $-6.598 \times 0.03 = -19.79\%$, and adding the convexity term gives

$$\frac{\Delta P}{P} \approx -6.598(0.03) + \tfrac{1}{2}(51.27)(0.03)^2 = -0.1979 + 0.0231 = -17.49\%,$$

while exact repricing at 4.50% gives \$82.32 per \$100, a fall of 17.68%. On \$50 billion the three estimates are losses of \$9.90 billion, \$8.74 billion, and \$8.84 billion: duration alone overstates a 300-basis-point loss by more than a billion dollars, and the convexity correction lands within \$0.1 billion of the exact figure, the same pattern Chapter 5 found for a single bond. The exact loss of \$8.84 billion is 98.2% of the bank's \$9 billion of equity. If the bonds are classified as held to maturity, none of it appears in the income statement or in regulatory capital, yet the bank's economic equity, its equity less the unrealized loss, is now $9 - 8.84 = \$0.16$ billion. Figure 8.3 sets the two balance sheets side by side.

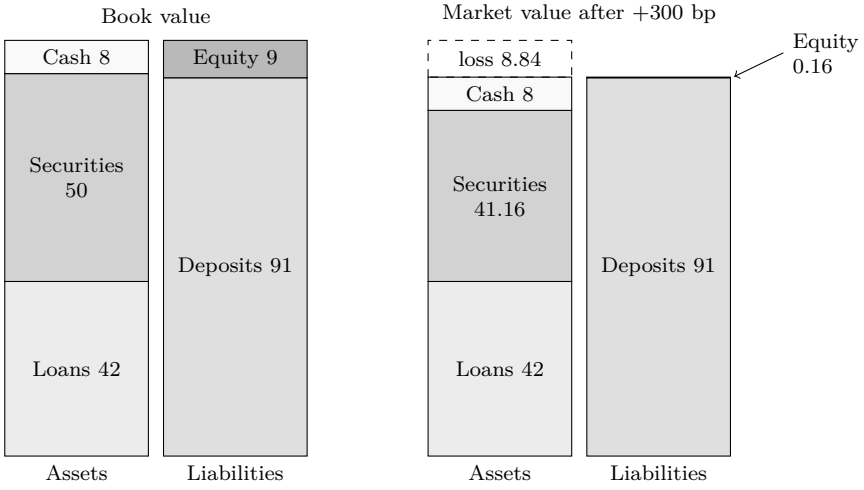


Figure 8.3: The stylized bank's balance sheet at book value and at market value after a 300-basis-point rise in rates, in \$ billions. The securities loss of \$8.84 billion leaves \$0.16 billion of economic equity, although the book-value balance sheet is unchanged.

EVE and the outlier test. Under a Basel-style parallel shock of +200 basis points, the assets' dollar duration is $50(6.598) + 42(0.25) = 340.41$, so the assets lose $340.41 \times 0.02 = \$6.81$ billion. What the deposits give back depends on their assumed duration. If the bank's deposit study says its depositors stay long enough to give the deposit base an effective duration of 3.5 years, their present value falls by $91(3.5)(0.02) = \$6.37$ billion, and Equation (8.10) gives $\Delta\text{EVE} = -(340.41 - 318.50)(0.02) = -\$0.44$ billion, 4.9% of Tier 1 capital and comfortably inside the 15% threshold of \$1.35 billion. If the deposits are flighty instead, with an effective duration of 0.5 years, they give back only \$0.91 billion and $\Delta\text{EVE} = -\$5.90$ billion, 65.5% of Tier 1 capital, a severe outlier. The balance sheet is identical in the two cases; one behavioral parameter has changed.

Net interest income. Over the next year at +300 basis points, the \$50 billion of cash and floating-rate loans earns an extra $50(0.03) = \$1.50$ billion. With a deposit beta of 0.25 the deposits cost an extra $0.25(91)(0.03) = \$0.68$ billion, and Equation (8.11) gives $\Delta\text{NII} = +\$0.82$ billion: on the earnings measure this bank *gains* from rising rates, at the very moment its economic equity has all but vanished. With a beta

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of 0.70, the kind of pass-through a bank faces once its depositors start moving money into money-market funds and Treasury bills, the extra deposit cost is \$1.91 billion and $\Delta\text{NII} = -\$0.41$ billion.

A run forces the loss into the open. Suppose a quarter of the deposits, \$22.75 billion, are withdrawn. Cash covers \$8 billion, so the bank must raise \$14.75 billion by selling bonds at 82.32% of par, which means selling 35.8% of the portfolio, \$17.92 billion of book value. The sale crystallizes 35.8% of the unrealized loss, \$3.17 billion, cutting book equity to \$5.83 billion with \$5.67 billion of loss still unrealized. Worse, selling out of the held-to-maturity category contradicts the stated intent to hold, which under U.S. accounting rules generally forces the rest of the portfolio to be reclassified and carried at fair value, so the remaining loss stops being invisible too. The bank's solvency did not change on the day of the run: its economic equity was \$0.16 billion before and after. The run removed the option to wait. And cash plus the entire securities portfolio covers only \$49.16 billion of withdrawals, 54.0% of deposits; beyond that the bank must borrow against its loans or fail.

Deposits, Runs, and Liquidity

The worked example makes the depositor the decisive variable, and three features govern how a deposit base behaves. The first is the deposit beta, which is not a constant: betas are low while rates are low and the gap between deposit rates and market rates is small, and they rise as that gap widens and depositors notice it. The second is the share of *uninsured deposits*, balances above the deposit-insurance limit, which in the United States is \$250,000 per depositor, per bank, per ownership category. An insured depositor has no reason to run, and an uninsured one every reason to run first: in the classic analysis of Diamond and Dybvig (1983), a bank that must liquidate long-term assets at a loss to pay early withdrawers cannot pay everyone, and the depositors who wait bear the shortfall. The third is how concentrated and connected the depositors are. A million households react to news independently and slowly; a few thousand firms advised by the same handful of investors can react together within hours. Mobile banking has removed the friction of the branch queue, and social media can carry the first signs of a run across an entire depositor network in an afternoon. Jiang et al. (2024) estimate that the rate rises of 2022 and early 2023 cut the market value of U.S. banks' assets by about 10%, roughly \$2 trillion, and show that losses of that size can expose even banks with liquid assets to self-fulfilling runs by uninsured depositors.

The Liquidity Coverage Ratio of Section 8.3.3 exists to absorb these risks. Its 30-day stress scenario assigns an outflow rate to each funding category, from 5% for stable, fully insured retail deposits to 40% for unsecured deposits from nonfinancial companies, and the bank must hold enough high-quality liquid assets to cover the result (BCBS 2013). But a 30-day calibration assumes a run unfolds over weeks. The run in the vignette below took less than two days.

8.3.6 Case Vignette: Silicon Valley Bank, March 2023

Silicon Valley Bank was the worked example at full scale. Its holding company, SVB Financial Group, grew from \$71 billion of assets in 2019 to \$211.8 billion at the end of 2022 on a flood of deposits from technology start-ups and the venture-capital firms that funded them (Board of Governors of the Federal Reserve System 2023; SVB Financial Group 2023). Lending could not keep pace, so the bank invested the money: at the end of 2022 securities made up 55% of its assets, against 25% at comparable large banks, and 78% of those securities were classified as held to maturity. The held-to-maturity portfolio, carried at \$91.32 billion, had a fair value of \$76.17 billion and a weighted-average duration of 6.2 years, an unrealized loss of \$15.15 billion against total stockholders' equity of \$16.0 billion, or 94.7% of it, with a further \$2.53 billion of unrealized losses in the AFS portfolio. About 94% of its \$173.1 billion of deposits were uninsured, against an average of 41% at its peers. The deposit base was already turning: noninterest-bearing deposits fell from \$125.9 billion to \$80.8 billion during 2022, and the bank's own model now assumed a deposit beta of about 70% on its interest-bearing deposits.

The measurement failures match the worked example too. The Federal Reserve's review found that the risk appetite set by the bank's board tracked only an NII metric, not EVE; that the EVE metric had breached internal limits for years; and that in April 2022 management responded by lengthening the assumed duration of its deposits, a change that shrank the measured mismatch without taking any risk off the balance sheet (Board of Governors of the Federal Reserve System 2023). That is the deposit-duration lever of the worked example, pulled in the direction that makes the outlier disappear. The review also found that under the rules

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in force before 2019 the bank would have had to count its AFS losses in regulatory capital, which would have been $1.9 billion lower, and to meet the full LCR, under which it would have been about 9% short of high-quality liquid assets in December 2022.

On 8 March 2023, SVB announced that it had sold $21 billion of AFS securities at a $1.8 billion after-tax loss and planned to raise $2.25 billion of new equity. Depositors read the announcement as a sign of distress. On 9 March they withdrew about $42 billion, 24.3% of year-end deposits, in a single day, leaving the bank with a negative cash balance of about $958 million (DFPI 2023). The Federal Reserve's review describes a run "fueled by social media" and by a concentrated network of venture investors and technology firms that withdrew their deposits together "with unprecedented speed"; for comparison, Washington Mutual, the largest bank failure of 2008, lost $19 billion of deposits over 16 days. Overnight, SVB told supervisors that it expected more than $100 billion of further outflows on 10 March, which together with the first day's $42 billion would have been 82.0% of year-end deposits within two days. The bank did not have the cash or the collateral to meet them. The California Department of Financial Protection and Innovation closed it on the morning of 10 March 2023 and appointed the Federal Deposit Insurance Corporation as receiver.

On 12 March, the Treasury, the Federal Reserve, and the FDIC invoked the *systemic risk exception* of U.S. deposit-insurance law, which allowed the FDIC to protect all of SVB's depositors, insured or not, and the Federal Reserve opened a new lending facility for banks secured by government securities valued at par rather than at market (U.S. Department of the Treasury et al. 2023). Losses to the Deposit Insurance Fund were to be recovered from the banking industry through a special assessment, which the FDIC's final rule of November 2023 set at an estimated $16.3 billion for the uninsured deposits protected at SVB and at Signature Bank, which had failed two days later (FDIC 2023).

Three lessons carry beyond this bank. First, the loss existed long before the run; the accounting deferred its recognition, and the run removed the option to wait. Second, the risk number was dominated by a behavioral assumption about depositors, and governance let that assumption move when the number breached its limit, the kind of model-risk failure Section 8.6.1 is meant to catch. Third, liquidity rules calibrated to a 30-day stress were not built for a run measured in hours. Early-warning models of deposit outflows, trained on daily client-level flows and on the network of who banks with whom, are a natural use of Chapter 6's classifiers, but SVB's record suggests that a missing signal was not the binding constraint: the EVE breaches were measured, and reported to a board committee, for years.

8.4 Sensitivity-Based Risk Limits

A desk can sit inside every VaR limit it has been given and still carry the position that ends its year. The reason is structural, not a failure of calibration: VaR compresses a book into one number, and the compression hides where the risk sits. Two desks reporting the same $300,000 daily VaR can hold portfolios with nothing in common, one hedged against the coming move and the other pointed into it.

Practitioners therefore run sensitivities alongside VaR, never instead of it. A sensitivity answers a much narrower question: not "how bad could tomorrow be, all things considered," but "if this one risk factor moves by one unit, and nothing else changes, how much do we make or lose?" The answers are unglamorous, linear, and immediately actionable, which is why the trading floor lives on them.

8.4.1 The Greeks as Limits in Their Own Right

Chapter 5 computed delta, gamma, vega, and theta for this chapter's option, and the delta-normal approximation used the first two as ingredients of VaR. On a derivatives desk each is a limit in its own right, monitored independently of VaR. Return to the market maker short 1,000 of Chapter 5's one-year $50-strike calls ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$), and read the Greeks as a risk report rather than pricing output. Table 8.4 collects them, scaled to the position and joined by rho, the one Greek Chapter 5 did not compute.

A risk manager reads the table line by line. The position is short delta, so it loses money if the stock rallies. It is short gamma, meaning that delta itself moves against the position in both directions, the effect the delta-normal example isolated numerically. It is short vega, so it loses if implied volatility rises. It is long theta, collecting $10.47 a day in time decay as long as nothing happens, the market maker's business model and the compensation for being short gamma and vega. (Chapter 5 quoted this option's theta as −$3.82 per

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Greek	Measures exposure to	Position	Limit	Utilization
Delta	a \$1 move in the stock	−611.5	1,500	40.8%
Gamma	the change in delta per \$1 move	−25.55	50	51.1%
Vega	a one-percentage-point move in implied volatility	−\$191.62	\$250	76.6%
Theta	the passage of one day	+\$10.47	—	—
Rho	a one-percentage-point move in interest rates	−\$237.00	\$400	59.3%

Table 8.4: The full Greek profile of a short position in 1,000 one-year \$50-strike calls, with representative desk limits

year; a risk report is read every morning, so desks quote theta per day, and dividing by 365 recovers the number above.) And it is short rho, though at this maturity that is the least of its concerns.

The key observation is one this chapter's VaR machinery cannot make. Every VaR figure computed for this position, delta-normal, delta-gamma, or full revaluation, shocks the *stock price* and holds implied volatility at 30%. The model is structurally blind to vega: implied volatility is not among the risk factors it perturbs, so no recalibration or backtesting will make that risk appear. Yet a five-point rise in implied volatility, from 30% to 35%, is an ordinary single-day move in a stressed market, and it would cost this position

$$1{,}000 \times \$0.1916 \times 5 = \$958,$$

against a 99% full-revaluation VaR of \$1,403.99 attributed to price moves alone. A routine volatility move, invisible to the risk report, is worth 68% of the entire price-driven 99% VaR. This is not an argument against VaR. It is an argument for reading a vega limit next to it.

8.4.2 Bucketed Sensitivities: The DV01 Ladder

The logic applies with more force to interest-rate books, where the risk factor is not a single price but a whole yield curve. The standard sensitivity there is DV01, the dollar value of a basis point: the profit or loss produced by a one-basis-point move in yields. A single aggregate DV01 for the whole book is the rates equivalent of a single VaR number, and it conceals the same thing.

Consider a desk whose positions are reported in the four standard maturity buckets of Table 8.5, with each figure giving the position's profit or loss from a one-basis-point *rise* in that bucket's yield, so that a long bond position appears as a negative number.

Bucket	DV01 (per bp)	Parallel +10bp	Flattening scenario
2-year	−\$25,000	−\$250,000	−\$500,000
5-year	−\$40,000	−\$400,000	−\$400,000
10-year	+\$38,000	+\$380,000	−\$190,000
30-year	+\$27,000	+\$270,000	−\$270,000
Net	\$0	\$0	−\$1,360,000

Table 8.5: A bucketed DV01 ladder whose net exposure is exactly zero

The net DV01 of this book is exactly zero. A report carrying only that aggregate would describe the desk as flat, and under a parallel shift it is: adding ten basis points to every yield produces gains and losses that cancel to the cent, as the middle column shows.

But the curve does not only move in parallel, and this desk is not flat. Its *gross* DV01, the sum of the absolute bucket exposures, is \$130,000 per basis point. It is long the short end and short the long end, a position that profits if the curve steepens and loses if it flattens. Applying a flattening scenario in which

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two-year yields rise 20 basis points, five-year yields rise 10, ten-year yields fall 5, and thirty-year yields fall 10 gives

$$(-25{,}000)(20) + (-40{,}000)(10) + (38{,}000)(-5) + (27{,}000)(-10) = -\$1{,}360{,}000,$$

a loss of $1.36 million from a curve move that the net-DV01 report scores as zero exposure. Figure 8.4 shows why: under the parallel shift the bucket gains and losses offset, while under the flattening every bucket loses at once. The bucketed ladder makes this visible, which is why rate desks have limits per bucket rather than one on the total. The principle transfers to equity and credit books: aggregate where you must, but never let the aggregate be the only thing anyone looks at.

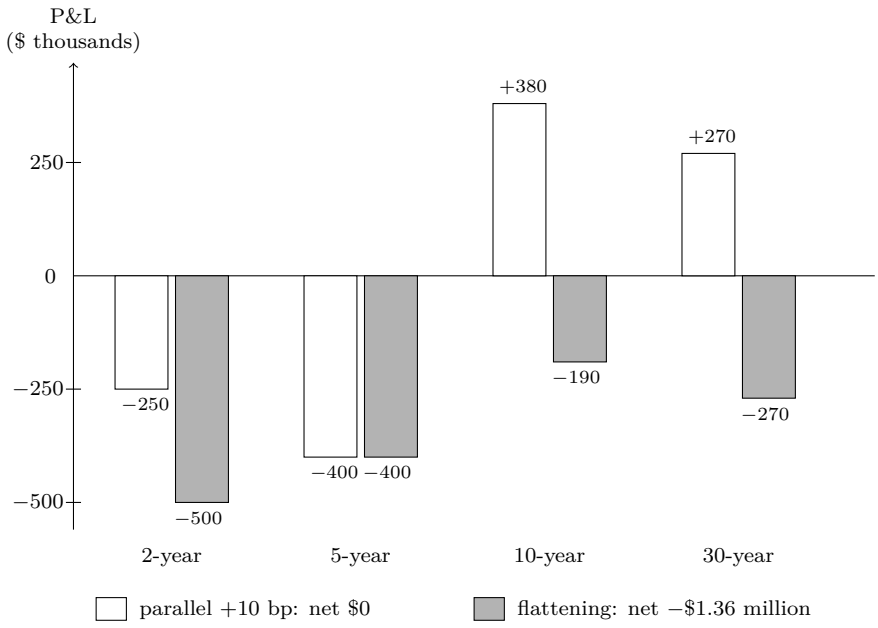


Figure 8.4: Bucket-by-bucket profit and loss of Table 8.5's ladder under a parallel +10 basis-point shift (open bars) and under the flattening scenario (shaded bars)

This is also the direction regulation has taken: the sensitivities-based method at the heart of the Fundamental Review of the Trading Book computes capital from these bucketed delta, vega, and curvature sensitivities rather than from a single portfolio-level loss distribution.

Chapter 7's principal component analysis of the yield curve explains *why* the ladder decomposes risk as it does. It found that almost all of the curve's daily variation is captured by three factors: a level factor that shifts all maturities together, a slope factor that tilts short against long, and a curvature factor. Projecting this desk's ladder onto those factors restates the whole example in one line. A net DV01 of zero is zero exposure to the *level* factor, which is why the parallel shift produces nothing; the desk's real position is a large exposure to the *slope* factor, and the flattening scenario is a slope-factor move. Bucketing by maturity is the practitioner's version of a factor decomposition, and a desk that has run the PCA can set limits on the three factor exposures directly, more compactly and more in line with how the curve moves than limits on four buckets treated as independent.

8.4.3 Computing Sensitivities at Scale: Adjoint Differentiation

The two preceding subsections asked for nine numbers, five Greeks and four DV01 buckets. A real trading book asks for thousands, across every instrument, risk factor, and maturity bucket, recomputed at least daily, and producing them is a computational problem with an elegant answer from machine learning.

The obvious method is to bump and reprice: value the book, add a basis point to one risk factor, value it again, and difference. It is easy and correct but does not scale, because each sensitivity costs a full revaluation.

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A book exposed to n risk factors costs $n+1$ valuations, and when instruments are priced by Monte Carlo, so that one valuation takes minutes, and n runs into the thousands, risk reporting becomes an overnight batch that a trader cannot interrogate during the day.

Adjoint algorithmic differentiation removes the dependence on n altogether. Its cost is bounded by a small constant multiple, typically three to five times, of a *single* valuation, no matter how many sensitivities are requested. A book that took an hour to produce one set of Greeks by bumping produces the entire set in minutes.

The mechanism is the one used to train neural networks. A pricing formula, like a network's forward pass, is a composition of elementary operations whose local derivatives are known analytically. Evaluating the formula once while recording that composition builds a computational graph. Seeding the output with a derivative of one and applying the chain rule backward through the graph then accumulates the derivative of the output with respect to every input simultaneously, in a single reverse sweep. This is reverse-mode automatic differentiation, and backpropagation is its application to a loss function instead of a price. The reason training a network with a million parameters costs only a constant multiple of a forward pass is the reason a bank can compute a thousand Greeks for the price of a few valuations: it is the same theorem.

Applying it to the Black-Scholes price of Table 8.4's option makes this concrete. One forward evaluation records a graph of roughly twenty elementary nodes; one reverse sweep then returns $\partial C/\partial S$, $\partial C/\partial \sigma$, $\partial C/\partial T$, and $\partial C/\partial r$ at once, giving a delta of 0.611539, a vega of 0.191623 per percentage point of volatility, a theta of -0.010472 per day, and a rho of 0.237003 per percentage point of rates. Each agrees with the closed form to nine decimal places, the method's other advantage: unlike a bumped finite difference, an adjoint derivative is exact up to floating-point rounding, with no step size to choose and no cancellation error to trade against truncation error.

Gamma is the instructive exception. It is a second derivative, and a single reverse sweep returns only first derivatives, so obtaining it requires differentiating the derivative, that is, applying the machinery to its own output. This is the distinction between a gradient and a Hessian in machine learning, and it carries the same cost implication: first-order sensitivities are nearly free, and second-order ones are not. Production systems handle this with nested automatic differentiation, the same construction used for Hessian-vector products when training a model with a second-order optimizer.

The technique entered finance through the work of Giles and Glasserman (2006) on Monte Carlo Greeks and is now standard on derivatives desks and in the valuation-adjustment calculations of modern derivatives infrastructure. It is not machine learning applied to finance in the usual sense, since nothing is learned from data. It is arguably more interesting: an algorithm popularized in one field answering a computational problem in the other, for reasons that lie in the structure of differentiation rather than the subject matter.

8.4.4 Case Vignette: Leveraged Gilts and the 2022 UK Pension Crisis

A UK defined-benefit pension fund owes its members payments stretching decades into the future, so the present value of its liabilities behaves like a very long bond: it rises when long-term interest rates fall. The natural hedge is to hold long-dated UK government bonds, gilts, whose prices move the same way. *Liability-driven investment* (LDI) is the strategy built on that hedge, and by 2022 more than £1 trillion was invested in it (Cunliffe 2022). Most funds wanted more interest-rate hedging than they had assets to spare, because they also held equities and other growth assets, so LDI managers supplied leverage: they bought gilts with money borrowed through repurchase agreements, or took the same exposure through swaps, against a cushion of the pension fund's own capital. Large schemes ran segregated mandates; smaller ones bought units in *pooled* LDI funds shared by many schemes. When gilt prices fell, the losses came out of the cushion and the banks on the other side of the repo and swap trades called for collateral in cash. The pension fund as a whole was usually no worse off, since its liabilities fell in value by more than its gilts did, but its LDI fund needed cash immediately.

On Friday 23 September 2022 the UK government announced a package of unfunded tax cuts, the "Growth Plan" or mini-budget. Long-dated gilt yields rose about 30 basis points that day, and on Monday the 30-year yield rose about 50 more, leaving it over 80 basis points above Friday morning's level. The Bank of England later told Parliament that the period saw two daily rises in the 30-year yield of more than 35 basis points, when the largest daily rise since 2000 had been 29, and that the four-day rise was more than twice the largest since 2000, set during the 2020 "dash for cash" (Cunliffe 2022). LDI managers reported that, if conditions

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kept worsening, they would have to sell at least £50 billion of long gilts into a market that traded about £12 billion a day. By the evening of Tuesday 27 September the 30-year yield was 67 basis points above that morning's level, and managers warned that several pooled funds would fall to negative net asset value and begin winding up the next day, with the banks that had lent to them selling the gilts held as collateral. On Wednesday 28 September the Bank announced temporary purchases of long-dated gilts, by daily auction until 14 October, and the 30-year yield fell by more than 100 basis points that day. It bought £19.3 billion in total, £12.1 billion of conventional and £7.2 billion of index-linked gilts, and sold all of them back by 12 January 2023 (Bank of England 2023).

The DV01 arithmetic of Section 8.4.2 explains the speed. Take a stylized 30-year gilt with a 3.5% annual coupon priced at par. Its modified duration is 18.39, nearly three times that of the bank's seven-year bonds in Section 8.3.5, and its convexity is 460.9. A single 35-basis-point day takes 6.16% off its price. Now put £1 billion of pension money into a pooled fund levered four times, holding £4 billion of these gilts. The fund's DV01 is £7.36 million per basis point, and one 35-basis-point day calls £247 million of collateral, 24.7% of the fund's capital, which must arrive in cash, typically by the next business day, from the trustees of many smaller schemes, few of whom could raise it that fast. A 100-basis-point rise, smaller than the four-day move of September 2022, calls £652 million, 65.2% of the capital, and a rise of 166 basis points wipes the fund out entirely; at two times leverage it would take 440 basis points. Convexity helps, since duration alone would have predicted a fall of 18.39% for a 100-basis-point rise against the exact 16.29%, but not nearly enough. In March 2023 the Bank's Financial Policy Committee concluded that LDI funds should withstand a yield rise of about 250 basis points at a minimum, on top of their buffers for ordinary daily moves (Bank of England Financial Policy Committee 2023). For this gilt a 250-basis-point rise takes 34.41% off the price, so the largest leverage that survives it is $1/0.3441 = 2.91$ times.

Three lessons connect this episode to the chapter. First, the hedge was economically sound and the risk was liquidity: the loss that mattered was the collateral call, not the scheme's funding position, so a VaR on its net exposure would have looked reassuring. Second, the danger was collective. Every leveraged fund held the same gilts and faced the same calls, so their selling raised yields and generated further calls, the fire-sale spiral that Section 8.3.3's liquidity-adjusted VaR, calibrated to one firm's spread, cannot see and Chapter 20's contagion models are built to study. Third, like Silicon Valley Bank's run, it happened in hours. A liquidity buffer sized to a 30-day horizon is irrelevant to a margin call due the next morning; the measure that matters is the yield move a fund can absorb with the cash it already holds, and that is a DV01 calculation.

8.5 Stress Testing and Machine Learning

Historical data cannot describe an event that has not happened yet, which is where scenarios and learned models both enter.

8.5.1 Stress Testing and Scenario Analysis

VaR and Expected Shortfall are backward-looking: even the parametric and Monte Carlo versions are calibrated to a historical mean and standard deviation, so they assume the future will resemble the recent past. Stress testing asks a different question: not "how bad are the worst outcomes the historical distribution suggests," but "what would happen to this portfolio under a specific, named adverse scenario," whether historical (such as replaying the return pattern from the 2008 financial crisis or the 2020 COVID-19 market shock) or hypothetical (such as a sudden 200-basis-point rate shock or a 30% equity market decline).

Applying a historical stress scenario, a single-day return of -7%, comparable to some of the worst individual trading days during the 2008 crisis, to the \$10 million portfolio in Table 8.2 gives a stressed loss of

$$\text{Stressed Loss} = 0.07 \times \$10{,}000{,}000 = \$700{,}000,$$

nearly three times the parametric 99% VaR of \$250,081 computed in Section 8.2.2.

The gap does not mean the VaR calculation was wrong; it is the point of stress testing. VaR describes a quantile of an assumed distribution, while a stress scenario explores an outcome a short, calm sample may never have contained. Regulators require large institutions to run such exercises (in the United States, the Federal Reserve's Comprehensive Capital Analysis and Review, or CCAR, is a prominent example)

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because a model calibrated on recent, calm data can understate the likelihood and severity of a crisis, the non-stationarity concern of Chapter 7.

Standard stress testing, as above, starts from a scenario and asks how much the firm would lose; reverse stress testing runs the same logic backward, starting from a specific outcome, such as the firm's capital falling below the regulatory minimum or the firm becoming insolvent entirely, and asking what combination of market moves, defaults, or funding shocks would be required to produce it. This counteracts a subtle bias in forward stress testing: designers drawing on recent history and their intuition tend to produce scenarios that resemble past crises, while reverse stress testing forces a search over the outcomes the firm cannot survive, whether or not they have ever occurred, and asks the risk committee to judge how plausible they are.

Suppose the desk running the $10 million portfolio of Table 8.2 has a risk capital cushion of $500,000 to absorb its trading losses before it needs capital from elsewhere in the firm. Reverse stress testing asks what single-day return would exhaust it. Since Loss $= |r| \times$ \$10,000,000, solving $500{,}000 = |r| \times 10{,}000{,}000$ gives $|r| = 5\%$, a single-day move of -5% or worse. Against the -7% historical scenario, the reverse test makes explicit what the forward scenario did not: a recurrence would not merely breach the cushion but exceed it by \$700,000 − \$500,000 = \$200,000, a capital shortfall rather than a limit breach, which a risk committee needs to know before such a day arrives.

Correlation Stress Testing: Quantifying "Correlations Go to One"

The case vignette below describes correlations rising sharply in a crisis; the two-asset position of Section 8.2.3 makes the claim numerical. Chapter 2's AssetA/AssetB covariance matrix implies a calm-period correlation of

$$\rho_{\text{calm}} = \frac{-0.000198}{\sqrt{(0.000414)(0.000118)}} \approx -0.896,$$

a strongly diversifying, negative relationship, which is why the equally weighted portfolio's volatility, 0.583%, is so much lower than either asset's own volatility ($\sigma_A \approx 2.03\%$, $\sigma_B \approx 1.09\%$) individually. Stress the correlation toward the crisis-period pattern described qualitatively in the vignette, holding each asset's own volatility fixed but replacing ρ_{calm} with a stressed $\rho_{\text{stress}} = 0.5$: the stressed covariance becomes $\rho_{\text{stress}}\sigma_A\sigma_B \approx 0.0001105$, and the resulting portfolio volatility is

$$\sigma_p^{\text{stress}} = \sqrt{w^\top \Sigma_{\text{stress}} w} \approx 1.372\%,$$

more than double the calm-period figure of 0.583%, so that $\text{VaR}_{95\%}^{\text{stress}} \approx 1.645(0.01372)(2{,}000{,}000) \approx$ \$45,141 (\$45,137 at full precision), roughly 2.35 times the calm-period parametric VaR of \$19,182 computed earlier, with neither asset's own volatility having changed at all. This is the quantitative content of the vignette's warning: a diversification benefit that looks large under calm correlations can shrink dramatically, and portfolio risk more than double, because assets that moved independently or oppositely start moving together, the mechanism that made 2008 more dangerous for diversified portfolios than their pre-crisis models suggested.

8.5.2 Machine Learning Applications in Market Risk

Every technique so far is a classical statistical model, deliberately, because a risk number feeding regulatory capital and executive decisions must be transparent and auditable. Machine learning nonetheless enters market risk management in several increasingly standard ways, each extending rather than replacing these tools.

Volatility forecasting is the most direct: a recurrent neural network or gradient-boosted ensemble can in principle capture nonlinear volatility patterns a fixed GARCH(1,1) cannot, feeding the volatility-adjusted VaR machinery a more flexible forecast, at the cost of the overfitting and interpretability concerns of Chapter 6.

8.5.3 A Head-to-Head Test: GARCH(1,1) versus Gradient Boosting on Asymmetric Volatility

Chapter 7 flagged a well-documented shortcoming of GARCH(1,1): its recursion depends only on ε_{t-1}^2, so it treats positive and negative shocks of the same size alike, when in practice negative shocks raise volatility more (the leverage effect). A gradient-boosted tree given the lagged return's sign as a feature can in principle learn this asymmetry, which plain GARCH(1,1) cannot represent.

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The companion notebook tests this directly: it simulates 750 days of returns from a known, asymmetric GJR-GARCH-style process ($\omega = 0.000005$, $\alpha = 0.05$, $\beta = 0.85$, with an extra variance term $\gamma = 0.12$ triggered only by negative shocks), fits a plain symmetric GARCH(1,1) by maximum likelihood on the first 600 days, and separately fits a gradient-boosted regressor on the same 600 days using lagged squared *and* signed returns as features, with hyperparameters chosen by time-series-aware cross-validation rather than an arbitrary guess.

Both models forecast one step ahead over the remaining 150 days, and because the true variance path is known (it is not observable in real markets, which is why the test needs simulated data), each forecast can be scored against it.

The result surprises only if one assumes flexibility always wins: the misspecified, symmetric GARCH(1,1) achieves a forecast-variance RMSE of 2.416×10^{-5}, beating the sign-aware, cross-validated gradient-boosting model's RMSE of 3.830×10^{-5} by 36.9%, even though the gradient-boosting model had direct access to the information (the shock's sign) that GARCH(1,1) cannot use. Gradient boosting still beats a naive constant-variance benchmark (4.202×10^{-5} RMSE) by 8.8%, so it learns something, but a correctly structured recursive model, propagating variance forward as the true process does, keeps a large advantage over a generic ensemble learning from lagged features, even one given the information it needs and carefully tuned. The result is useful and sobering: it is why volatility forecasting in practice augments GARCH-family models with richer inputs (implied volatility, order flow, cross-asset signals) rather than replacing the recursion with an off-the-shelf model trained on lagged returns.

A second application addresses a computational bottleneck. Full revaluation of a large options book becomes prohibitively slow once thousands of positions must be repriced at thousands of Monte Carlo scenarios, so some desks train a neural network as a fast pricing surrogate, learned once from a pricing model such as Black-Scholes-Merton or a full Monte Carlo engine and then queried at simulation speed.

A third, newer application uses generative models, the GANs and diffusion models of Chapter 16, to synthesize plausible stress scenarios beyond anything in a firm's historical window, addressing the limitation of Section 8.5.1: a sample that never included a severe crisis cannot suggest one, but a generative model of the structure of extreme moves can propose a novel, plausible one.

In each application the discipline is unchanged: a more flexible model is validated, backtested, and governed at least as rigorously as a classical one, because its flexibility makes an undetected flaw easier to introduce and harder to catch.

8.6 Governance and Managing the Risk

Measuring a risk is not the same as owning it; this part covers who validates the numbers and what is actually done about them.

8.6.1 Model Validation and Model Risk Management

Every method in this chapter, VaR, Expected Shortfall, GARCH-based volatility forecasting, liquidity adjustment, is a model in the sense of Chapter 1: a simplified representation of reality whose outputs are only as trustworthy as its assumptions. Model risk management governs this formally, subjecting every model used for a material financial decision to independent scrutiny before deployment and monitoring afterward, rather than trusting it because a competent team built it.

A mature model risk management framework, closely following supervisory guidance such as the U.S. Federal Reserve and OCC's SR 11-7 (superseded in April 2026 by the interagency SR 26-2, which keeps the same pillars (Federal Reserve, OCC, and FDIC 2026)), organizes this scrutiny around three activities. Independent validation requires that a model be reviewed by staff who did not build it, checking the soundness of its assumptions (does normality make sense for this portfolio's returns), its implementation (does the code compute what the specification says), and its outcomes (does backtesting, as in Section 8.3.1, confirm its performance). Ongoing monitoring tracks performance after deployment, since a model sound at launch can degrade as markets change, and catches the degradation before it causes a material loss. Model inventory and documentation keep a current record of every model, its owner, validation status, and known limitations, since a firm that does not know what models it has, or where they are used, cannot govern the risk they pose.

In practice, a firm with hundreds of models cannot validate all of them with equal intensity, so most model risk management frameworks tier models by materiality. A model whose output could plausibly affect more

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than $50 million of capital or reported risk might be classified Tier 1, requiring full independent validation and annual revalidation, while a narrowly scoped model with less than $5 million of potential impact might be Tier 3, revalidated only every two to three years. This chapter's portfolio VaR model, given its role in regulatory capital, would sit in the highest tier at any bank of meaningful size.

The same governance applies to Chapter 9's credit models, and it is what separates a well-run risk function from a collection of clever models. The methods of this chapter are necessary but not sufficient, since even a correct model can mislead dangerously when applied outside the conditions under which it was validated, the failure mode of this chapter's 2008 case vignette.

8.6.2 Managing the Risk, Not Just Measuring It

Every tool so far answers "how much could we lose?" None answers the question a desk head must act on: "what are we going to do about it?" A risk report that never changes a position is an expensive newsletter; measurement earns its keep when a number crosses a threshold and someone trades differently.

A firm can do four things with a risk it has measured. It can **accept** the risk, which is a legitimate decision when the exposure is small relative to capital and is being paid for. It can **reduce** it by cutting the position. It can **hedge** it by taking an offsetting position in a related instrument, keeping the exposure it wants and shedding the exposure it does not. Or it can **transfer** it to someone else through an insurance contract or a derivative, which Chapter 11 develops from the perspective of the party accepting such risks for a living. The rest of this section works through the two middle options, which require calculation.

8.6.3 Hedging: Removing an Exposure You Do Not Want

Return once more to the short call position of Section 8.4. Its delta of -611.5 says that the position loses $611.50 for every $1 the stock rises. A market maker who wants the option's time decay without a directional view eliminates that exposure by buying 611.5 shares. The hedged book's delta is zero, and the first-order price risk is gone.

What is left is the interesting part. Delta hedging removes the linear exposure, but the position is still short 25.55 units of gamma, the rate at which the hedge goes stale as the stock moves. At the 99% one-day move of $2.198 used in the delta-normal example, the delta-hedged position still loses

$$\tfrac{1}{2} \times 25.55 \times (2.198)^2 = \$61.72,$$

compared with the $1,403.99 full-revaluation loss unhedged. The hedge removes 95.6% of the 99% loss, a large improvement but not a complete one. Two features of the residual matter more than its size. It is quadratic, so doubling the shock roughly quadruples it. And it is *one-directional*: a short-gamma position loses whether the stock rises or falls, because the sign of ΔS disappears when squared. The hedge has converted a directional bet into a bet that the stock will not move much, a different risk rather than none.

That is the general shape of hedging: a hedge exchanges a risk you do not want for a smaller, differently shaped one you are willing to hold, and the residual is what the risk report must track.

The residual is also the opening for a machine learning treatment, which Chapter 19 applies to this position. The calculation above assumes the hedge can be rebalanced continuously and for free. It cannot: every adjustment to the share position pays a bid-ask spread, so a hedger who rebalances often enough to keep delta near zero pays away the option premium in transaction costs, while one who rebalances rarely enough to keep costs down carries the residual gamma risk just computed. There is no closed-form answer to where the optimum sits, because it depends on the cost structure, the volatility, and how much tail risk the desk is willing to tolerate. Chapter 19 solves it as a sequential decision problem, finding a rebalancing band that beats daily delta hedging on both average cost and tail risk simultaneously, and then generalizes the band to a neural hedging policy trained directly on the distribution of outcomes. The classical delta hedge computed here is the baseline that work is measured against.

8.6.4 Position Sizing: Controlling Risk Before It Exists

Hedging acts on an existing position. The cheaper control acts before the position is put on, choosing its size so that the resulting risk is the risk the firm wanted.

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Volatility targeting is the standard mechanical version. Take this chapter's \$10 million portfolio, whose daily return standard deviation of 0.937% was computed in Section 8.2.2. Scaling to an annual figure in the conventional way gives

$$\sigma_{\text{annual}} = 0.937\% \times \sqrt{252} = 14.87\%.$$

Suppose the mandate this portfolio is run under specifies a 10% annualized volatility target. The portfolio is running roughly half again as much risk as authorized, and the fix requires no view on any holding: scale the whole book by the ratio of target to realized volatility,

$$\frac{10\%}{14.87\%} = 0.672,$$

holding \$6.72 million in the risky portfolio and the remaining \$3.28 million in cash. Because volatility scales linearly with position size while the composition is unchanged, this hits the target exactly, without the estimation difficulties of Chapter 12's mean-variance optimizer.

The catch makes position sizing a forecasting problem rather than arithmetic. The scaling factor is a ratio of a target to an *estimate*, so the control is only as good as the estimate, and volatility is not constant, which is what Chapter 7's GARCH models capture. Sizing on a trailing standard deviation rather than a conditional forecast that responds to yesterday's shock is the difference between de-risking after a turbulent week and de-risking during it. Section 8.5.3 compares GARCH(1,1) with a gradient-boosted tree on a series with a known asymmetric volatility response, and the winner is the model that should drive the scaling factor. Volatility targeting turns a better volatility forecast directly into a better-sized position.

A volatility-targeted portfolio rescales continually, cutting exposure as markets become turbulent and adding it back as they calm. That is intended, but it means selling into falling markets mechanically, alongside every other volatility-targeted fund doing the same arithmetic on the same day, a mechanism for the correlated forced selling that Section 8.5.1 stress-tests for.

8.6.5 Stop-Losses and Drawdown Triggers, and Why They Disappoint

The last common control is the crudest: when losses reach a threshold, cut the position, whatever any model says. Stop-losses appeal because they need no distributional assumption and cannot be gamed by a misspecified covariance matrix.

They also underdeliver, as this chapter's data shows. Section 8.2.5 computed a maximum drawdown of 6.74% for the twenty-day portfolio, reached on the final day. Suppose the desk had operated a rule liquidating the portfolio whenever its drawdown from the running peak reached 5%. Tracking the drawdown day by day, the rule would not have fired until day 19, and liquidating at that point would have avoided only the final day's −1.49% return. Out of a 6.74% peak-to-trough decline, the stop-loss rescues about 149 basis points.

The reason is generic. A drawdown trigger lags by construction: it cannot fire until the loss has been taken, so it concedes the whole distance to the threshold plus whatever happens before the fill. In a series that recovers, the same rule crystallizes a loss that would have reversed, at the point of maximum pessimism. Stop-losses are a last-resort control against catastrophic tails and operational failure, not a technique that improves expected outcomes. The tools that reduce risk systematically are sizing the position correctly beforehand and hedging unwanted exposures.

8.6.6 Risk Governance: The Three Lines of Defense

Every method in this chapter exists inside an organizational structure that determines whether it is used correctly, and financial institutions typically organize risk oversight with a "three lines of defense" model, illustrated in Figure 8.5.

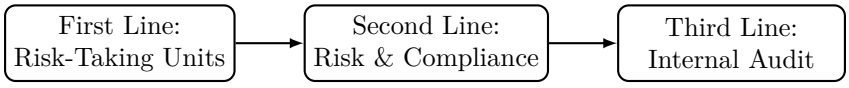


Figure 8.5: The three lines of defense in risk governance

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The first line of defense is the business unit that takes the risk, a trading desk or a lending team, and is responsible for managing it day to day within approved limits. The second line is an independent risk management and compliance function, organizationally separate from the business units it oversees, that builds and maintains models like the VaR and economic capital methods introduced in this chapter and the credit-scoring methods of Chapter 9, sets risk limits, and monitors whether the first line is operating within them. The third line, internal audit, periodically and independently verifies that the first two lines function as intended, checking not just whether a number was computed correctly but whether the whole process was followed. The structure prevents the single point of failure that arises when the people earning revenue from a position also measure and report its risk, a conflict implicated in many historical risk management failures.

A model is only as trustworthy as the governance around it: a perfectly specified VaR model computed by a risk function with no real independence from the desk it monitors offers little protection. The second line is typically led by a Chief Risk Officer (CRO) with direct access to the board's risk committee, independent of the chief executive and business-line management, so that a concern can reach the board without being filtered by the managers it reflects on. This independence is not a formality. Several well-documented failures trace back to a risk function that identified a problem but lacked the standing to have it acted on before it caused a material loss, which is why governance codes and regulators treat CRO independence, board-level risk reporting, and the second line's authority to veto or unwind a position as essential.

A limit example shows how the three lines interact day to day. Suppose the second line has approved a daily 99% VaR limit of \$300,000 for the desk running the \$10 million portfolio from Table 8.2. The parametric 99% VaR of \$250,081 computed in Section 8.2.2 represents a limit utilization of $250{,}081/300{,}000 \approx 83.4\%$, well within the limit, so the first line trades normally and the second line's monitoring is routine. Now suppose realized volatility rises sharply enough, by the volatility-adjusted VaR logic above, that the next day's VaR is \$340,000, a breach of \$40,000, or roughly 13.3% over the limit. This is the event the three lines exist to catch. The breach is reported to the second line the same day, which assesses whether it reflects a temporary spike in volatility or a change in the desk's risk profile that warrants an immediate reduction. A breach that recurs or is not promptly remediated is what the third line's audit is designed to flag, even if the second line's daily response was adequate.

8.7 Case Vignette: When Value at Risk Meets a Genuine Crisis

The 2008 financial crisis is the standard cautionary tale about the limits of this chapter's methods, and the failure was not one of arithmetic. Many large banks entered 2008 with VaR models built and validated as this chapter describes: parametric or historical methods, calibrated on several years of historical returns, backtested against the kind of exception-counting procedure in Section 8.3.1, and generally passing those backtests during the calm years leading up to the crisis. The most famous symptom came early, in the August 2007 "quant quake" that preceded the main crisis. Explaining heavy losses at Goldman Sachs's own quantitative hedge funds, the firm's chief financial officer, David Viniar, told the *Financial Times*: "We were seeing things that were 25-standard deviation moves, several days in a row." A normal distribution gives a single 25-standard-deviation move a probability so small that it should not occur once in the history of the universe, let alone on several days in a row.

The arithmetic was not wrong; the normality assumption of Section 8.2.2 was a poor description of returns in a crisis, the fat-tails caveat of this book. A model calibrated on a multi-year window without a severe crisis will underestimate extreme moves, since it has never observed one, the finite-sample problem behind this chapter's caution about short backtesting windows. Moreover, correlations between asset classes that had looked low or stable rose sharply, a pattern summarized as "correlations go to one in a crisis," so the diversification assumed by portfolio VaR, including the variance-covariance approach of Section 8.2.3, protected far less than historical correlations suggested.

The lesson is not that VaR or Expected Shortfall are poor tools; both remain required practice at every major institution. It is that each method answers its question only as well as its assumptions hold, so stress testing, backtesting, and the judgment of an independent second line are necessary complements to any single statistical model.

The regulatory response shaped the landscape this chapter describes. The Basel Committee's Fundamental Review of the Trading Book (FRTB), published in January 2016 and revised in January 2019 (BCBS 2016b), but still only partly in force in October 2026 (Table 8.6), replaced VaR with Expected Shortfall as the basis

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for market-risk capital calculations. ES, unlike VaR, is sensitive to the severity of losses beyond the threshold, not only their frequency, which addresses the "25-standard-deviation event" problem: a VaR-based charge is blind to how much worse the tail is, an ES-based charge is not. FRTB also requires calibration to a period of significant historical stress rather than only recent, possibly calm data, and introduces desk-level model approval so that a bank-wide validation can no longer mask weaknesses in one desk's risk measurement. None of this makes the chapter's methods obsolete; it holds them to a more demanding, crisis-aware standard than the pre-2008 VaR framework.

Four FRTB mechanics go beyond what this chapter builds. First, the internal-model measure is Expected Shortfall at 97.5%, not 99%: under normality the two give almost the same number, since the 99% VaR multiplier is $z_{0.99} = 2.326$ while the 97.5% ES multiplier is $\phi(z_{0.975})/0.025 = 2.338$, so the change of confidence level keeps the capital level roughly where it was in calm, Gaussian conditions and charges more only where the tail is fatter than normal. Second, the single 10-day horizon of the old rules gives way to liquidity horizons of 10, 20, 40, 60, or 120 days by risk factor, so that a position in an illiquid credit spread is capitalized over the time it would actually take to exit. Third, a desk keeps internal-model approval only if it passes, alongside backtesting, a profit-and-loss attribution test showing that the risk model's own P&L tracks the front office's pricing P&L closely. Fourth, risk factors without enough real price observations to model, the non-modellable risk factors, are carved out of the ES model and charged a separate stress-scenario add-on. National implementation has lagged the standard badly, as Table 8.6 records.

Jurisdiction	FRTB status as of October 2026
Basel standard	Published January 2016, revised January 2019; Basel implementation date 1 January 2023
Canada, Japan	In force since November 2023 (Canada, the first jurisdiction) and March 2024 (Japan's banks)
Singapore, Hong Kong	In force since the start of 2025
European Union	Own-funds requirements postponed twice by delegated act, to 1 January 2027; a June 2026 delegated act, still under scrutiny by Parliament and Council, adds a temporary capital multiplier and operational relief for 2027–2030
United Kingdom	Standardized approach from 1 January 2027 with the rest of Basel 3.1; internal model approach from 1 January 2028
United States	2023 proposal withdrawn; re-proposed 19 March 2026 (comments closed 18 June 2026); no final rule

Table 8.6: Implementation of the FRTB market-risk standard by jurisdiction, as of October 2026 (BCBS 2020; European Commission 2025; European Commission 2026; PRA 2026; Board of Governors of the Federal Reserve System et al. 2026)

8.8 Challenges in Market Risk Management

Several challenges recur across nearly every market risk management application, and Chapter 9 revisits several of them in a credit-risk setting.

Model risk in the risk model itself. Every method in this chapter, parametric VaR's normality, the GARCH recursion's parametric structure, the delta-normal method's linearity, rests on assumptions that can fail when they matter most, in a crisis, because crises are unusual events a model calibrated on typical conditions was not designed to capture. The gap between a linear approximation and full revaluation, small for a modest move, widens when the move is largest.

Correlation and tail dependence. Individual risk measures can look reasonable in isolation, but a portfolio becomes dangerous when many positions move together at the worst time, a pattern called tail dependence. The case vignette showed it: correlations that looked low and stable rose sharply in 2008,

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eroding the diversification the variance-covariance VaR of Section 8.2.3 assumed. Estimating tail dependence directly, rather than assuming a stable correlation matrix, is statistically difficult and highly consequential.

Procyclicality. Risk measures calibrated on recent data tend to show low risk during calm periods (encouraging more risk-taking and leverage) and high risk immediately after a shock has already occurred (forcing deleveraging precisely when the system can least afford it), an amplifying feedback loop implicated in several financial crises that no purely statistical measure resolves.

Data quality and limited loss history. Severe crashes are fortunately rare, so the data for estimating their probability and severity are limited, the small-sample problem illustrated by this chapter's twenty-observation VaR and backtesting examples.

Regulatory and business tension. Risk models live in an organization that wants to grow revenue: a risk function that is too conservative can be overridden or under-resourced, one that is too permissive can miss a crisis, and the balance is as much a governance challenge as a statistical one.

Regulatory arbitrage. Whenever capital is calculated from a bank's own reported VaR, as under the Basel multiplier, the bank has an incentive to minimize the capital charge rather than the risk, by choosing a favorable window, confidence level, or specification that backtests well. The gap between measured and true risk can widen quietly even when every calculation is correct.

Limit-setting on a diversification-sensitive measure. Component VaR's dependence on the book's correlation structure cuts both ways. It rightly rewards diversifying positions with a smaller charge, but a desk near its limit has an incentive to seek positions that look diversifying under calm-period correlations, the positions Section 8.5.1's correlation stress test warns can start compounding losses when correlations move against the book in a crisis.

Machine learning adds a further layer of model risk. A neural network or gradient-boosted model can in principle forecast volatility or tail risk more flexibly than GARCH or a historical window, but it inherits Chapter 6's overfitting and interpretability concerns where an undetected error is unusually costly. An opaque VaR model that a regulator or validator cannot interrogate is hard to trust with a bank's capital requirement, whatever its backtested accuracy.

8.9 Key Takeaways

Market risk management complements the prediction-focused methods of earlier chapters by asking how bad outcomes can get, not just what is expected to happen. Value at Risk and Expected Shortfall summarize the downside of a return distribution by parametric, historical, or Monte Carlo methods, each trading distributional assumptions against finite-sample noise. Expected Shortfall is preferred because it accounts for the severity of losses beyond the threshold, though backtesting severity is harder than counting VaR exceptions.

Volatility need not be fixed: GARCH lets VaR update immediately after a large move, filtered historical simulation adds the historical method's distribution-free shape, and the delta-normal approximation understates options risk in proportion to the gamma it ignores, a gap the delta-gamma correction nearly closes.

Component VaR decomposes total risk into per-position contributions that depend on the book's correlation structure, which is why pushing correlations toward the crisis pattern can more than double measured risk with no position's own volatility changing.

No aggregate should be the only thing anyone looks at. Sensitivities sit alongside VaR because a single number discards *where* the risk is: a rate book can report a net DV01 of exactly zero while carrying a $1.36 million exposure to a curve flattening, and an options position's VaR can be computed correctly to the cent while remaining structurally blind to a routine move in implied volatility. Producing those sensitivities at scale is solved by reverse-mode automatic differentiation, the algorithm that trains neural networks.

The banking book carries the same interest-rate risk under a different accounting. A bond portfolio held to maturity is not marked to market in earnings or, for many banks, in regulatory capital, but its loss is real: a 300-basis-point rise took 17.68% off the stylized bank's seven-year bonds and left it $0.16 billion of economic equity out of $9 billion, while its net-interest-income measure showed a gain. Economic value of equity and net interest income measure the two sides of that exposure, and both rest on behavioral assumptions about depositors, their beta and effective duration, that alone can turn a supervisory outlier into a comfortable pass.

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Silicon Valley Bank showed what happens when concentrated, uninsured depositors test those assumptions: a loss that had been building for a year was forced into the open in two days.

Measurement is not management. Hedging exchanges an unwanted exposure for a smaller, differently shaped residual; position sizing controls risk before it is taken and is only as good as its volatility forecast; and stop-losses lag by construction and rescue far less of a drawdown than intuition suggests. None of these is self-policing: backtesting (including Basel's traffic-light zones), independent validation, and governance such as the three lines of defense are what catch a miscalibrated model before it causes a loss.

As in Chapter 5, a risk number is only as trustworthy as its assumptions, and sound practice combines several methods and an account of where they disagree rather than relying on any single measure. Chapter 9 now turns to the analogous toolkit for credit risk.

8.10 Suggested Exercises

8.1 Explain, in your own words, the difference between Value at Risk and Expected Shortfall, and describe a situation in which two portfolios could have identical VaR but very different Expected Shortfall.

8.2 Using the returns in Table 8.2, compute the parametric 90% VaR (using $z_{0.90} = 1.282$) for the \$10 million portfolio, and compare it to the 95% and 99% figures computed in the chapter.

8.3 Using the historical method, compute the 90% VaR and Expected Shortfall from Table 8.2 (following the convention of Section 8.2.2, which reads the 95% VaR from the second-worst return, leaving one observation beyond it, the 90% VaR leaves two observations beyond it and so corresponds to the third-worst return out of twenty).

8.4 Using Table 8.2's cumulative value path and the definition in Section 8.2.5, compute the drawdown at Day 6 (relative to the running peak established by that point) and compare it to the chapter's maximum drawdown of 6.74%, realized only at Day 20. Explain why a portfolio can pass through a large intermediate drawdown that is later exceeded by an even larger one, rather than the path recovering.

8.5 Using Chapter 7's GARCH(1,1) recursion values at $t = 2$ ($\sigma = 2.03\%$) and $t = 5$ ($\sigma = 1.99\%$), compute the parametric 95% VaR of a \$5 million position at each date, and explain why the two figures are so much closer together than the $t = 3$-versus-$t = 4$ comparison in the text.

8.6 Explain why a portfolio's 99% VaR can be smaller than its loss under a stress-test scenario, and why this is not evidence that the VaR calculation is wrong.

8.7 A bank's VaR model produces 7 exceptions over a rolling 250-day window at the 99% confidence level. Using the Basel traffic-light framework, classify this outcome and explain what supervisory consequence it would typically trigger.

8.8 Using this chapter's delta-normal example, recompute the 95% and 99% delta-normal and full-revaluation VaR for a position that is instead *long* 1,000 calls, and explain which direction of stock-price move now represents the position's worst case.

8.9 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause a risk management system to understate a firm's true risk.

8.10 Using Section 8.2.3's two-asset covariance matrix, recompute the portfolio's volatility and 95% VaR under a stressed correlation of $\rho = 0.8$ instead of the $\rho = 0.5$ used in the text, and compare the resulting increase to the one computed in Section 8.5.1.

8.11 A bank has \$60 million in high-quality liquid assets and projected net cash outflows of \$55 million over a 30-day stress scenario. Compute its LCR, and explain why a bank could satisfy the LCR comfortably while still having an inadequate NSFR.

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8.12 A third asset, AssetC, is added to Section 8.2.3's two-asset position with a Component VaR of \$4,000. Using the fact that Component VaRs must sum to the portfolio total, explain what this implies about how AssetC is correlated with the existing AssetA/AssetB position, without needing to know AssetC's own standalone volatility.

8.13 Using this chapter's three-lines-of-defense limit example, compute the limit utilization if the approved daily VaR limit were instead \$275,000 (holding the parametric 99% VaR of \$250,081 fixed), and explain whether a subsequent rise in VaR to \$290,000 would count as a limit breach under this tighter limit.

8.14 Using this chapter's reverse-stress-testing example, compute the single-day return that would exhaust a risk capital cushion of \$800,000 instead of \$500,000 for the same \$10 million portfolio, and explain whether the -7% historical stress scenario from Section 8.5.1 would still exceed this larger cushion.

8.15 Explain, using this chapter's FRTB discussion, why a capital framework based on Expected Shortfall is less vulnerable than a VaR-based framework to a repeat of the "25-standard-deviation event" problem described in the 2008 case vignette.

8.16 Using the square-root-of-time rule from this chapter's economic-capital discussion, compute the 5-day regulatory VaR implied by the parametric 99% VaR of \$250,081, and explain why this figure is smaller than the 10-day figure computed in the chapter even though both are scaled from the same 1-day VaR.

8.17 This chapter's economic-capital discussion showed that a modest positive autocorrelation of $\phi = 0.10$ makes the square-root-of-time rule understate 10-day VaR by about 9.4%. Explain, in your own words, why a *negative* autocorrelation (mean reversion) in daily returns would instead make the square-root-of-time rule *overstate* multi-day risk, and describe a market condition in which returns might plausibly exhibit this pattern.

8.18 Using the vega in Table 8.4, compute the profit or loss on the short 1,000-call position if implied volatility rises by three percentage points while the stock price does not move at all, and express it as a percentage of the position's 95% full-revaluation VaR of \$980.79 from the delta-normal example. Explain why no amount of backtesting the VaR model would have revealed this exposure.

8.19 Apply a *steepening* scenario to the DV01 ladder of Table 8.5, in which two-year yields fall 10 basis points, five-year yields fall 5, ten-year yields rise 10, and thirty-year yields rise 15. Compute the desk's profit or loss, and explain why the sign of the result is the opposite of the flattening scenario worked in the text even though the net DV01 is zero in both cases.

8.20 A trading book is exposed to 500 distinct risk factors. (a) How many full valuations are required to produce a complete set of first-order sensitivities by bump-and-revalue? (b) Roughly how many are required by adjoint algorithmic differentiation, and why does this figure not depend on the number 500? (c) Explain why gamma, unlike delta, is not obtained for free in the same reverse sweep, and name the machine learning computation that faces the same first-order-versus-second-order cost distinction.

8.21 Using Section 8.6.4's volatility-targeting calculation, recompute the dollar amount to hold in the risky portfolio (a) if the mandate's target volatility were 15% rather than 10%, and (b) if the daily volatility estimate were instead Chapter 7's GARCH post-shock forecast of 2.08%, holding the 10% target. Comment on the size of the position change in (b) and on what it implies about how often a volatility-targeted portfolio must trade.

8.22 Show that the residual loss on a delta-hedged short-gamma position grows quadratically, by recomputing Section 8.6.3's figure of \$61.72 for a stock move twice as large as the \$2.198 used there. Explain why the residual is a loss for *both* a rise and a fall in the stock price, and what this implies about the kind of bet a delta-hedged option seller is really making.

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8.23 Using the stylized bank of Section 8.3.5: (a) estimate the loss on its securities portfolio from a 200-basis-point rise in yields, first with duration alone and then with duration and convexity, and express each as a percentage of the bank's \$9 billion of equity; (b) find the effective deposit duration at which the +200 basis point ΔEVE of Equation (8.10) exactly reaches the supervisory outlier threshold of 15% of Tier 1 capital, and discuss what your answer implies about how much of the outlier test's verdict rests on the bank's deposit study rather than on its balance sheet.

8.24 For the same bank: (a) find the deposit beta at which the +300 basis point change in net interest income from Equation (8.11) is exactly zero; (b) recompute the securities sale, the realized loss, and the remaining book equity if 40% of deposits, rather than 25%, are withdrawn; (c) explain why a board that monitors only net interest income, as Silicon Valley Bank's did, can approve a balance sheet that is close to economic insolvency.

8.25 Repeat Section 8.2.8's Student-t calculation with $\nu = 6$ instead of $\nu = 4$. (a) Compute the scale s and the 99% VaR and Expected Shortfall of the \$10 million portfolio with Equation (8.6), and compare them with Table 8.3. (b) Find the confidence level at which the $\nu = 6$ VaR crosses the normal VaR, and compare it with the 97.47% crossing for $\nu = 4$. (c) Compare the $\nu = 4$ and $\nu = 6$ VaR at 95% and at 99.9%, and explain why the choice of ν, which twenty days of data cannot settle, matters little for the first and a great deal for the second.

8.26 For the stylized 30-year gilt of Section 8.4.4: (a) find the yield rise that wipes out a pooled LDI fund levered three times; (b) compute the collateral call on a fund with £1 billion of capital levered four times when the 30-year yield rises 67 basis points, the move between the morning and the evening of 27 September 2022, both exactly and with duration alone; (c) compute the largest leverage that survives a 250-basis-point rise using duration alone, compare it with the exact 2.91 times, and explain which way the duration approximation errs and why this error is conservative for a long bond position.

8.27 **Challenge (real data).** Download at least two years of daily prices for a small portfolio of real stocks (for example, three to five components of a major index, via `yfinance`) and compute the portfolio's daily returns using fixed weights of your choosing. (a) Compute the portfolio's parametric and historical 99% one-day VaR, as in Section 8.2.2, and compare the two estimates. (b) Backtest your parametric VaR estimate over the full sample using Kupiec's test, referenced in Section 8.3.1: count the number of days on which the realized loss exceeded the VaR estimate, and determine whether the exception rate falls inside the Basel traffic-light green zone for your sample size. (c) Real portfolio returns are typically both fat-tailed and volatility-clustered, unlike the twenty-observation toy example this chapter builds on; discuss whether your backtest results suggest the parametric (normal-distribution) VaR model over- or understates risk relative to the historical method, and connect this to this chapter's own discussion of GARCH-based, volatility-adjusted VaR.

8.11 Further Reading

- Jorion (2007), *Value at Risk*. The standard reference for VaR methodology, covering the parametric, historical, and Monte Carlo approaches introduced in this chapter in much greater depth and rigor.
- Hull (2023), *Risk Management and Financial Institutions*. A broad treatment of market, credit, and operational risk within financial institutions, extending this chapter's introductory coverage of each risk type.
- McNeil et al. (2015), *Quantitative Risk Management*. A mathematically rigorous treatment of coherent risk measures, Expected Shortfall, and the tail-dependence and correlation issues raised in this chapter's challenges section.
- BCBS (2010), *Basel III: A Global Regulatory Framework for More Resilient Banks and Banking Systems*. The primary regulatory source for the capital adequacy, backtesting, and liquidity-coverage concepts introduced in this chapter. Its market-risk successor, BCBS (2016b), *Minimum Capital Requirements*

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for Market Risk, is the standard produced by the Fundamental Review of the Trading Book: it replaced Value at Risk with Expected Shortfall as the regulatory measure and introduced the sensitivities-based capital charge that this chapter's treatment of sensitivity-based risk limits develops.

- Giles and Glasserman (2006), "Smoking Adjoints: Fast Monte Carlo Greeks." The paper that introduced adjoint differentiation to derivatives risk management, and the direct source for Section 8.4.3's claim that a full set of sensitivities costs a small constant multiple of a single valuation.
- Baydin et al. (2018), "Automatic Differentiation in Machine Learning: A Survey." A clear and self-contained account of forward- and reverse-mode automatic differentiation, written for the machine learning audience, making explicit the equivalence with backpropagation that Section 8.4.3 relies on.
- Crouhy et al. (2014), *The Essentials of Risk Management*. A practitioner-oriented synthesis spanning market, credit, and operational risk, model validation, and risk governance, tying together the full breadth of topics covered in this chapter and Chapter 9 in a single volume.
- BCBS (2016a), *Interest Rate Risk in the Banking Book*. The Basel standard behind Section 8.3.5's supervisory outlier test, setting out the six standardized rate-shock scenarios, the treatment of non-maturity deposits, and the disclosure requirements for EVE and NII sensitivity.
- Board of Governors of the Federal Reserve System (2023), *Review of the Federal Reserve's Supervision and Regulation of Silicon Valley Bank*. The Federal Reserve's own account of Silicon Valley Bank's failure and of the supervisory response to it, and the primary source for most of the vignette in Section 8.3.6, including the bank's EVE limit breaches and its change to the assumed duration of its deposits.
- Jiang et al. (2024), "Monetary Tightening and U.S. Bank Fragility in 2023: Mark-to-Market Losses and Uninsured Depositor Runs?" Marks the whole U.S. banking system's assets to market after the 2022 rate rises and models how unrealized losses and uninsured deposits combine into run risk, generalizing the worked example of Section 8.3.5 from one stylized bank to the entire U.S. banking system.

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